PROBABILISTIC SEISMIC HAZARD ASSESSMENT OF SANKHUWASABHA DISTRICT

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DECLARATION

This is to certify that the work which is being presented in the thesis entitled "**PROBABILISTIC SEISMIC HAZARD ASSESSMENT OF SANKHUWASABHA DISTRICT"** in partial fulfillment of requirement for the award of degree of Master of Engineering in Earthquake submitted in the Post Graduate Department of Earthquake Engineering at Khwopa Engineering College, Bhaktapur under Purbanchal University, Nepal. The matter presented in this thesis has not been submitted by me in any other university/institute for the award of ME in Earthquake.

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ABSTRACT

Seismic risk in the form of imminent disaster has been seen from the past records that the moderate to large earthquakes have caused the loss of life and property in Nepal. Hence, seismic hazard assessment for district level is essential and logical. Probabilistic Seismic Hazard Assessment for Sankhuwasabha District has been carried out. Sankhuwasabha district lies in the geological boundary of 27°06'N 86°57'E to 27°55'N 87°40'E. Information on tectonic setting, seismicity and prediction relation have been reviewed for Sankhuwasabha district.

Entire study area is divided into 200km x 200km grid zone and seismicity within 600km square was considered. Historical earthquake as well as instrumental earthquake from various source has been collected, all data has been converted to moment magnitude, repeated events and aftershocks have been removed. Completeness test was performed based on (Stepp, 1972) and Guttenberg – Richter parameters were determined based on the distribution of earthquake for each source zones.

With the comprehensive combined catalog and Ground motion Prediction Equation proposed for subduction zone by (Youngs et al., 1997) has been used to calculated the hazard of Sankhuwasabha district. Mean hazard curves has been developed for Sankhuwasabha district for rock site and soil site condition.

Results are presented in the form of Peak Ground Acceleration (PGA) and uniform hazard spectra for two different return periods 475 and 2475 years for both rock and soil sites.

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1. INTRODUCTION

1.1 Background

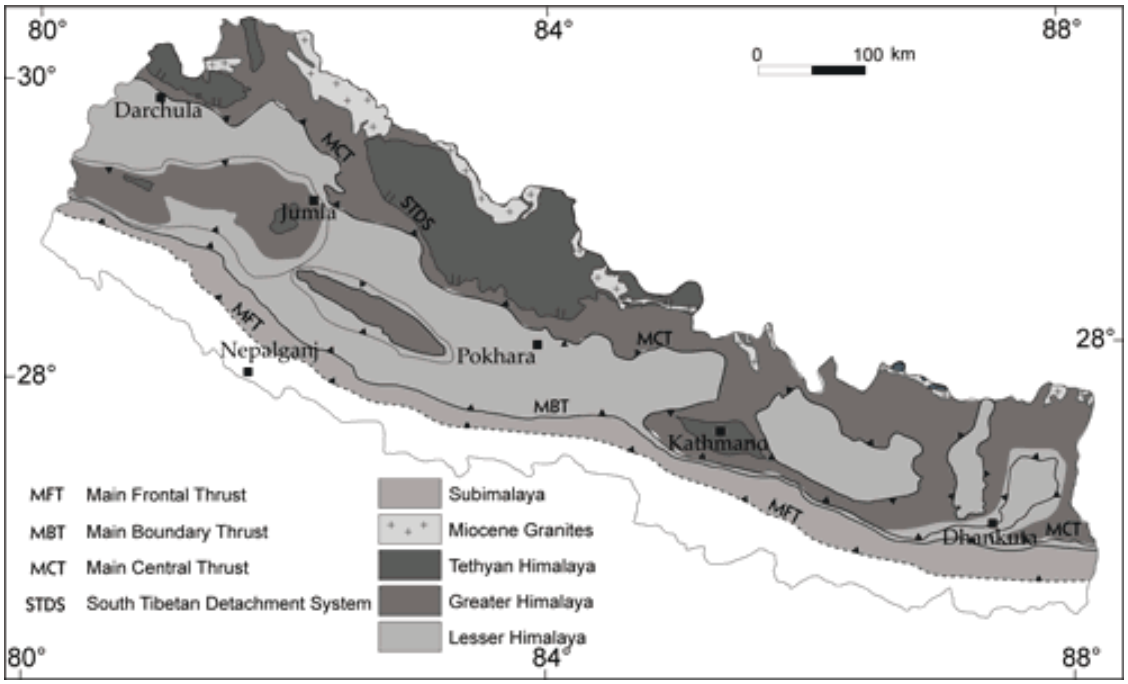
Nepal is an area of high seismic hazard, with the potential for large earthquakes up to Mw 9.2 (Stevens & Avouac, 2016). Earthquake is a naturally occurring event which is capable of causing death, injuries and damages in terms of losses of lives and property. On average, 10,000 people die each year due to structural and non-structural damage caused by earthquakes, while annual economic losses are in billions of dollars and often a large percentage of the gross national product of the country is affected (Elanashai & Sarno, 2008). To reduce this loss from the impending earthquakes, the task of earthquake engineering professionals is to work for earthquake resistant design of structures. Hence, it is felt that for every region, field-based design ground motion parameters should be available during the time of analysis, design and construction of earthquake resistant engineering structures.

In Nepal, moderate to great earthquakes may occur due to the fragile Himalaya region. It is required to do some level of seismic hazard analysis prior to vulnerability assessment of any area. Specially, for eastern region of Nepal, a seismic hazard analysis has not been carried out in detail yet. Apart from construction of National pride projects like Arun III hydropower projects and cities with highly growing industrial areas like Biratnagar, Dharan and Itahari no any detail hazard analysis has been done (Bhattarai G. , 2010). According to Disaster Risk Management Profile (2005), about 725 people were killed and 65000 houses were damaged due to 1988 earthquake (M6.6), most of them were from Dharan. It can be an important warning of seismic risk for impending disaster. Therefore, this thesis work is aimed to perform seismic hazard analysis considering most of the probable potential earthquake sources to achieve hazard curve, and hazard spectra which will be useful for dynamic analysis, earthquake resistant design, liquefaction hazard, seismic risk analysis and disaster management planning the potential of causing great damages.

Nepal lies towards the southern collisional boundary of the Indian and the Eurasian Plate, occupying the central sector of the Himalayas, nearly one-third of the 2,400 km long Himalayas Le Fort (1996). Within the narrow width of Nepal, three fault systems, Main Central Thrust (MCT), Main Boundary Thrust (MBT) and Himalayan Frontal Thrust (HFT), pass east to west throughout the length of Nepal. HFT bounds the terai plain tectonic area to the north. Terai plain is composed of alluvial deposits of depth of about 1500m. Churia range, composed basically of sandstone is located to the north of terai plain and is bounded by MBT and HFT. Mahabharat hills and midland can be categorized as Lesser Himalayan (LS) tectonic area, which is bounded by MBT and MCT. This area is mainly composed of schist, and granite in few parts. Higher Himalaya zone including Tethyan Himalaya tectonic area is bounded to the south by MCT.

Along the sides of these three greater fault systems and in the Tibetan Himalayan region, 92 small faults have been identified (BECA, 1993). The convergence of two plates is 19mm per year (Jouanne, et al., 2004), which has caused many earthquakes in the past in excess of moment magnitude M8. This indicates high rate of deformation and seismicity in the region. Four great earthquakes, in 2015 with M7.8, in 1934 with M8.1 (Rana, 1934), in 1833 with M7.7 (Bilham, 2004) and in 1505 with M8.2 (Ambraseys & Jackson, 2003) have been observed in Nepal Himalayas, which has resulted in tremendous loss of life and property.

To reduce or mitigate such type of loss from upcoming earthquake, the main task of earthquake engineering professionals is to work for earthquake resistance design of structures. Hence, an attempt is made for seismic hazard analysis of Sankhuwasabha district.



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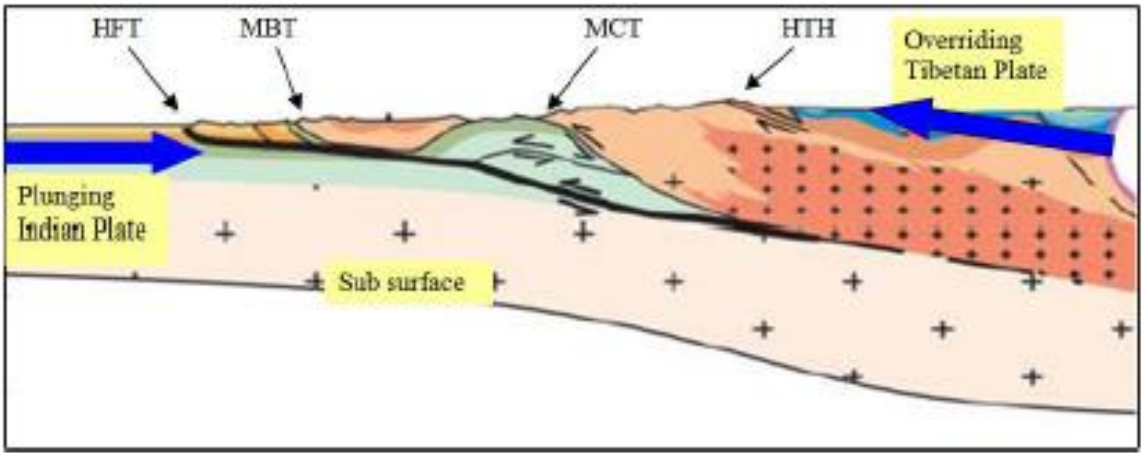


Figure 1.1.2 North-South cross section of Nepal through Kathmandu showing depth and faulting system after Avouac, (2003)

1.2 Rationale of the Study

Study on seismicity in Nepal Himalaya shows remarkable variation in seismic activities from one place to other in terms of micro seismic activity. In addition to other common factors, effect of earthquake depends on the local geology and site conditions. Currently available code in practice used for earthquake resistant design of building clusters different regions into a uniform single zone. Which is not reasonable to generalize different seismic sites under one seismic region without appropriate detail study of hazard level of individual area. In order to accurately estimate earthquake loadings in structures during earthquake, site dependent specific study of seismic hazard and ground response analysis is necessary.

With the available generalized data regarding seismic response, structure constructed in Nepal are seismically vulnerable. National Pride project, Arun-III hydropower plant is being developed on the Arun River of Sankhuwasabha District of Province I, East Nepal. It is and export-oriented project with a power generation capacity of 900 MW. Sankhuwasabha district have huge resources for development of such big projects but no works have been done yet for estimation of seismic hazard. Therefore, this work is dedicated to study probabilistic seismic hazard due to earthquakes that are overdue nearby this area.

1.3 Objective of the Study

General Objective

The primary objective of the study is to perform seismic hazard assessment for Sankhuwasabha district.

Specific Objective

The secondary objective of the study is:

1. Development of a computer program for carrying out PSHA.
2. Development of Guttenberg Richter relationship for each source zone.
3. Exploration of suitable attenuation relationship from existing ones.
4. Computation of Seismic hazard curve and prepare uniform hazard spectra.

1.4 Significance of the study

Nepal Himalaya shows remarkable variation in seismic activities from one place to another in terms of micro seismic activity. Effect of earthquake depends on source properties, path characteristics and local site effects. Earthquake resistant design code divides different regions into a uniform single zone which is not reasonable to categorize different seismic site under one seismic region without study of hazard level of individual area. Site dependent specific study of seismic hazard and ground response analysis is necessary to accurately estimate earthquake loading in structure during earthquake.

Sankhuwasabha district has a huge resource for the potential development of National level hydropower projects including fast growing cities which are seismically vulnerable because of generalized building design practice. Currently no work has been done yet for estimation of seismic hazard in this part of nation. This work is, therefore dedicated to study seismic hazard assessment due to earthquakes that are overdue nearby this district.

This research work includes development of "Probabilistic Seismic Hazard Analysis Software Module", a computer program developed in Excel/Visual Basic for Applications (VBA) to implement Probabilistic Seismic Hazard Analysis (PSHA) calculation process. Reusable code capability of this application can help other interested researchers to implement custom user defined functions and methods regarding PSHA. This will significantly reduce the time-consuming process of probabilistic hazard calculation process.

1.5 Limitations of the study

Despite the several efforts during the limited time frame given by the university, this study has given useful results to some extent objectives are fulfilled. However, there are few shortcomings, which can be considered in future researches to come up with more precise and applicable findings. The main shortcomings of this study are number of earthquake sources i.e. only Nine source zones have been considered. Other limitation can be prediction in constants in Guttenberg and Richter Recurrence relationship and the selection of attenuation relationship. The deaggregation in PSHA has not been carried out only due to the limitation of the time frame and scope of the present work.

1.6 Organization of the study

This report is organized in five sections the includes the objective, approach, methodology, results and conclusion of the work in this dissertation to identify and characterize seismic hazard of Sankhuwasabha district.

Chapter 1, Introduction, includes the background, objectives as well as limitation of the current thesis work. It also includes background information regarding the seismic history of Nepal and study area which set the stage for the need for performing seismic hazard assessment of Sankhuwasabha district.

Chapter 2, Literature Review, provides evaluative summary of studies found in the literature related to PSHA, various methods for declustering earthquake, completeness analysis and selection of attenuation relation for subduction zone including geology and tectonic setting of Nepal and Himalayan region that are used to define and characterize the seismic source were studied.

Chapter 3, Methodology, provided the step by step information for conducting PSHA. This section also explains the uncertainties.

Chapter 4, Presentation and Analysis of Data, this chapter includes the example of detail calculation of one source zone, which includes G-R relation, Hazard curve, Uniform Hazard Spectra.

Chapter 5, Conclusion and Recommendations, lists the major conclusion of the study on the basis of results drawing from the study.

2. LITERATURE REVIEW

2.1 Probabilistic Seismic Hazard Assessment (PSHA)

Probabilistic seismic hazard analysis (PSHA) is the most widely used approach for the determination of seismic design loads for engineering structures. The uses of probabilistic concept allow prediction of probability occurrence of a certain level of ground shaking at a site by considering uncertainties in the size, location, rate of occurrence of earthquake, and the predictive relationship (Kramer, 2007). In addition, PSHA provides a framework in which these uncertainties can be identified, quantified and combined in a rational manner to provide a more complete picture of the seismic hazard.

The expert panel on Seismic Hazard Analysis (1988, Washington DC) has outlined the method, use and importance of seismic hazard analysis. It states "The purpose of probabilistic seismic hazard analysis (PSHA) is to evaluate the hazard of seismic ground motion at a site by considering all possible earthquakes in the area, estimating the associated shaking at the site, and calculating the probabilities of these occurrences."

McGuire & Robin (2004) writes on his monograph that "PSHA and risk analysis are input into a larger decision making". He defines seismic hazard curve as the frequency plot of ground motion amplitude greater than specified value. He has accepted that Poisson model of ground motion occurrences can give the satisfactory results for probability calculation of general civil engineering structures.

Pandey et al. (2002) has performed Probabilistic Seismic Hazard Analysis to prepare seismic hazard map of Nepal by using commercial software "CRISIS 99" prepared by Institute de Ingenieria UNAM, Mexico. They have divided whole region of Nepal into ten Aerial sources and twenty four linear sources with approximately 40km length of each. For the purpose of calculation of peak ground acceleration they have used attenuation relationship propose by (Youngs, et al., 1997). Their work is limited within the PGA on bed rock. They have left the work about amplification of ground motion due to geological condition or local soil effect. Hence, in this thesis work site specific hazard curves and spectral curves are calculated and plotted for both hard rock and soft soil sites.

Parajuli et al. (2008) study gives the probabilistic hazard estimate throughtout Nepal considering historical earthquakes, intra plate slip and faults. Also a typical case probabilistic spectra is plotted for Pokhara city. For the city, design earthquakes for three probabilities of exceedance are simulated which can be useful to design new structures and retrofit of existing structures.

Khatiwada (2009) has prepared seismic hazard map of Nepal. In his report the maximum level of shaking in Eastern Nepal at bed rock level is 0.65g. He has performed probabilistic seismic hazard analysis considering ten aerial sources defined by National

Seismological centre, Nepal. In his work he has not considered the linear sources proposed by National Seismological Centre, Nepal.

Bhattarai (2010) has performed Probabilistic Seismic Hazard Analysis of Biratnagar Sub-Metropolitan city, Eastern Nepal. He has determined the Peak Ground Acceleration (PGA) and uniform hazard spectra for two different return periods of nearly 500 and 2500 years for both rock and soil sites of Biratnagar. The PGA of 0.29g is obtained at the rock site. The maximum spectral acceleration for return period of 500 years is calculated as 0.50g at rock site.

Bhattarai et al. (2011) a probabilistic seismic hazard assessment has been carried out taking 1934 Bihar-Nepal and 1988 Udayapur earthquake as point source. Although this prelimary study has some limitations on nature of seismic sources, the study is able to reflect the level of seismic hazard in the region. The Peak Ground Acceleration values for 1934 Bihal-Nepal earthquake varies from 100 to 350 gal at bed rock level.Similarly, for the 1988 Udayapur earthquake Peak Ground Acceleration varies from 100 to 300gal.

Thapa & Wang (2013) has estimated seismic ground motion hazard for Nepal using a probabilistic approach. Earthquake catalog from 1255 to 2011 has been compiled. According to catalogued earthquakes, together with available geological and tectonic information were used to delineate 23 seismic sources zones in Nepal and the surrounding regions. The estimated PGA values are in the range of 0.07-0.16g, 0.21-0.62g, and 0.38-1.1g for 63%, 10%, and 2% probability of exceedance in 50 years, respectively. The resulting ground motion maps show different characteristics of PGA distribution, i.e. high hazard in the far-western and eastern sections, and low hazard in southern Nepal. The quantified PGA values at bedrock level provides information for micro-zonation studies in different part of the country.

Chaulagain et al. (2015) has done seismic risk assessment and hazard mapping of Nepal. Study estimates the higher seismic hazard in the mid-western and eastern part of Nepal, whereas southern Nepal has the lowest seismic hazard. The expected PGA at 10,5,2 and 1% probability of exceedance in 50 years is found in the range of 0.22-0.50, 0.30-0.64, 0.42-0.85, and 0.51-1.07g. The results show that Biratnagar sub-metropolitan city, Kathmandu metropolitan city, and Pokhara sub-metropolitan city have spectral acceleration of 0.70, 0.67 and 0.66g respectively for 475 years return periods.

2.2 Earthquake Catalog

Historical (pre-instrumental) earthquake catalogues within study area were collected from published literatures. Data of historical earthquake is available for strong earthquakes only which caused considerable destruction in the past.

Ambraseys & Douglas (2004) presented list of significant earthquakes in northern India (15° -36°N,65°-95°E) from the year 1411 to 1974 which are the events collected from (Ambraseys & Jackson, 2003) and (Ambraseys & Bilham, 2003a). Earthquake information from 1411 to 1842 were collected from this catalogue.

Ojha et al. (2013) has published a catalogue for Nepal for the area between 20° to 35°N and 78° to 92°E for the period between 1255 AD to 2012 AD. They have divided the earthquake catalog into historical catalog (1255-1910) and instrumental catalog (1911-2012).

Rajaure & Paudel (2018) has prepared a comprehensive earthquake catalogue for Nepal and its adjoining region. The earthquake catalogue contains homogenized independent earthquakes of magnitude (Mw) between 4.0 and 8.5, occurred between 1100 AD and 2018 AD. The catalogue contains date, time, latitude, longitude, depth and magnitude of earthquakes, which are primarily required in the seismic hazard analysis.

Instrumental catalog used in this work consists of earthquakes reported by US geological survey department (USGS) and International Seismological center (ISC).

2.3 Magnitude Conversion

The composite earthquake catalogue contains earthquakes collected from various sources, which have reported different types of magnitudes (e.g., moment magnitude (Mw), Body wave magnitude (Mb), Surface wave magnitude (Ms) and Local magnitude (Ml). Moment magnitude is generally considered the best overall estimate of an earthquake size and does not saturate even at large earthquakes (Kanamori, 1977), (Hanks and Kanamori, 1979), therefore it is preferred in the study of seismic activity, seismotectonic and seismic hazard.

In order to develop homogenous catalog of consistent magnitude, the other types of magnitudes (M_s, M_b, M_L) were converted into moment magnitude (Mw) using empirical relations.

Ambraseys (1990) presents uniformly calculated surface-wave magnitude of over 400 earthquakes in the European area and adjacent regions associated with strong-motion recording. He derived the relationships between body wave, local magnitude and surface magnitude. So that by unifying the magnitude it can be used with attenuation law.

For this thesis work the different magnitudes were homogenized to moment magnitude M_w using empirical relation given by (Scordilis, 2006).

2.4 Declustering

Aftershock are earthquake events which are connected with parent event. It is a most important step of PSHA to remove Non-Poissonian behaviour from earthquake catalog, since PSHA relies on the assumption that seismicity follows a Poisson process. After a large earthquake, numerous aftershocks occur at near space and time whose occurrence is dependent on the appearance of the main shock (Musson, 1999). Declustering removes depended earthquakes (i.e., foreshocks and aftershocks) so that the retained earthquakes follows a Poisson distribution. Catalog of main shocks can be used in estimating seismic risk by virtue of statistical model when aftershocks are removed from total event listing.

Gardner & Knopoff (1974) algorithm, is a windowing method of declustering which is currently used by various researchers as well as U.S. Geological Survey. It is based on magnitude dependent spatial and temporal windows. Accordingly, the detection of offspring events is solely depends on the size of the windows. It fails to reflect the changes in background rates that have occurred in the past few years due to removal of 80% of earthquakes (Teng & Baker, 2019). Due to the window method implemented in Gardner & Knopoff algorithm this method frequently identifies aftershocks with much stronger shaking intensities than their associated mainshocks.

The linked-window method (Reasenberg, 1985) variant of (Gardner & Knopoff, 1974) windowing method, a specified distance interval L(M) and a specified time interval T(M) have been defined for each earthquake with a magnitude of M. Subsequent earthquakes will be identified as an aftershock if they occur within this time-space window. Foreshocks will be identified the same as aftershocks. The largest shock in a sequence has the largest time-space window and is a main-shock. Compared to the Gardner & Knopoff algorithm, this method is able to capture the changing hazard level in the region.

2.5 Completeness Test

Stepp (1972), Analysis of completeness of the earthquake sample in the Puget Sound area and its effect on statistical estimates of earthquake hazard gives the method of completeness analysis.

Earthquake data are not uniformly distributed, only few records are available in early periods and numbers of records have been increasing towards end. All earthquake catalogs are biased against small shocks, because of seismograph station density or, in the early records, population density. It is necessary to obtain the interval of complete reporting in each intensity class over which that class is homogeneous. A separate mean rate of occurrence can then be determined from above interval for each intensity class. In order to do completeness analysis (Stepp, 1972) of data, events need to be grouped into small interval of time and each magnitude ranges need to be judged separately. Slope changes in the graphs indicates changes of the completeness of the catalog. It is commonly assumed that the most recent change in slope occurs when the data became complete for each intensity class (Gasperini & Ferrari, 2000).

Tinti & Mulargia (1985) has proposed a visual cumulative method which is most common and simple method used for the completeness. It is a simple, graphical procedure based on the observation that earthquakes follow a stationary occurrence process. For a given magnitude class, the period of completeness is considered to begin at the earliest time when the slope of the cumulative fitting curve can be well approximated by a straight line.

2.6 Gutenberg-Richter relation

The Gutenberg-Richter relation is an empirical relation between the magnitude (M) of earthquake and number (N) of earthquakes with magnitude larger than 'm'. It is a remarkable feature of worldwide seismicity (Warner & Laura, 2003) which is used to describe seismic activity (Takeo & Kunihiko, 2008) and (Gutenberg & Richter, 1944) proposed the relation in the following form.

$$\log\lambda = a - bm \qquad (2.1)$$

Where, λ is mean annual rate of exceedance of magnitude M,

10^a is mean yearly number of earthquakes of magnitude greater than or equal to zero.

b is a constant describing relative likelihood of large and small magnitude earthquakes. When the b value increases, the number of large earthquakes decreases compared to the number of smaller magnitudes and vice versa. For active tectonic regions b value is about 1 and for volcanic regions it is up to 2 or higher. The a and b value are fundamental seismic parameters required in probabilistic seismic hazard analysis. The b value is estimated using least square method or the maximum likelihood method.

2.7 Ground motion attenuation relations

There are many ground motion attenuation relations that have been developed for different conditions. (Youngs et al., 1997) has been developed from worldwide seismic environment including (Crouse, 1997) catalogue. (Atkinson & Boore, 2008) compiled the database of (Youngs et al., 1997) and (Crouse, 1997) and developed new ground motion relation with four times big database for subduction zone. (Gregor et al., 2002) relation has been also developed for Cascadia subduction zone. (Youngs et al., 1997) , (Gregor et al., 2002), (Atkinson & Boore, 2008) and (Zhao, et al., 2006) can be used for subduction zone.

(Maskey & Mishra, 2005) has concluded that among the different attenuation relationships prepared for different regions their suitability depends not only on one law but also in magnitude range and source to site distance. For example, where attenuation relationship cannot be developed due to lack of complete set of earthquake catalogue, it is accepted to use attenuation relationships proposed by (Youngs et al., 1997), Donovan and (Cornell, 1968) which give way the PGA values nearer to the values equivalent to the recorded intensities during the past earthquakes of 1833, 1934 and 1988.

In Nepal different researchers has used different attenuation law during their study. (Bhattarai et al., 2011) has used the attenuation law proposed by (Cornell et al., 1979) in his study seismic hazard assessments for eastern Nepal using 1934 and 1988 earthquake. (Thapa & Wang, 2013) has used the attenuation relationship developed by (CEA, 2005) for western China. The elliptical attenuation model of PGA for the major axis and minor axis can be expressed as (CEA, 2005).

In subduction zones, there is equal possibility of occuring both interface as well as intraplate earthquake. Past earthquakes in Himalaya regions have never been categorized as interface or intra-plate earthquake. With the available information in the region, there is shallow angle thrust faults which is similar of subduction interface earthquakes as in the other part of the world. Intra-plate earthquakes are basically categorized by deep focus and volcanic activities. Himalaya of Nepal has no any reported evidences of volcanic activities. So, subduction interface ground motion relations are considered in this thesis work.

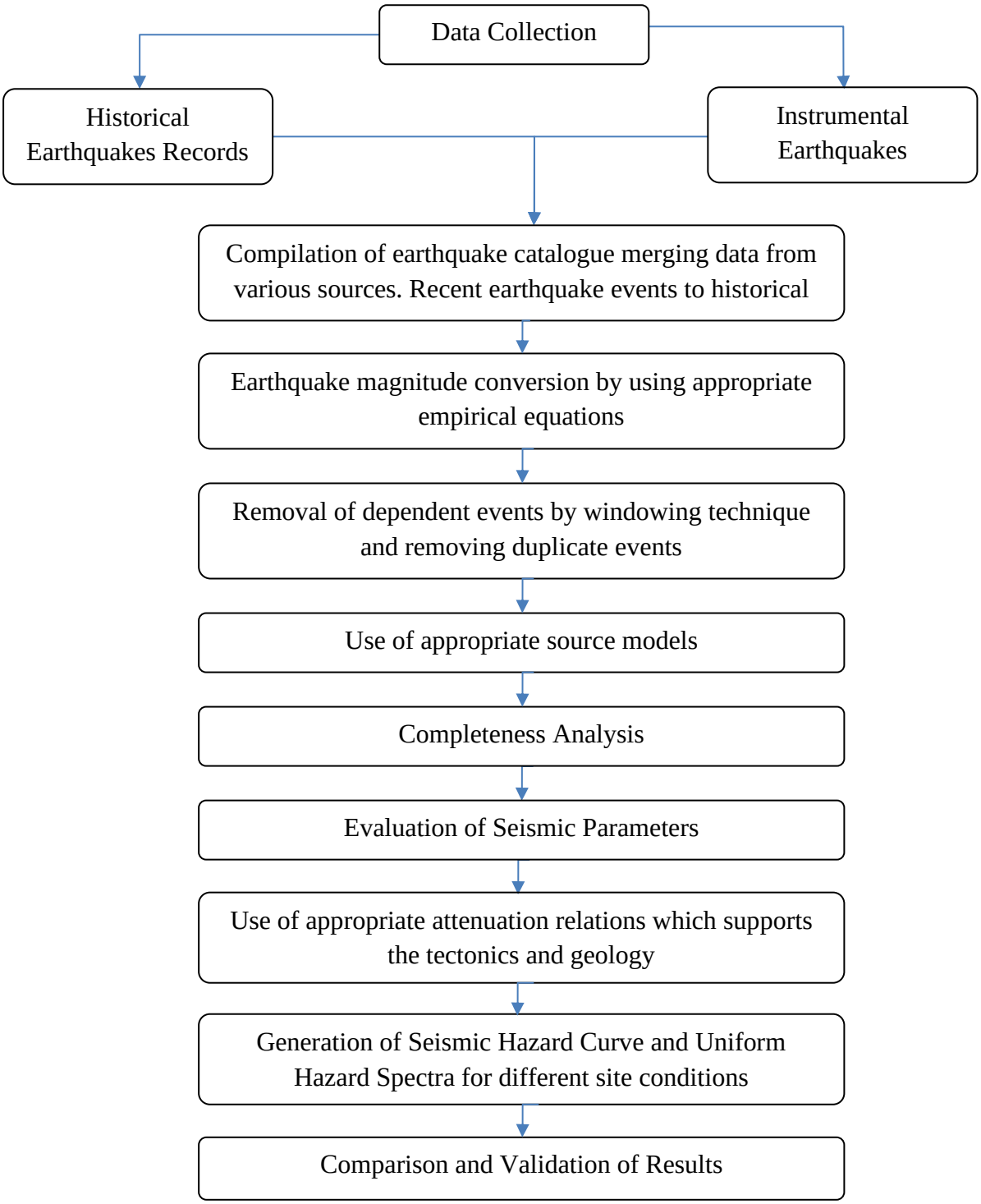
Among different attenuation relationship available Youngs et. al. 1997 have been selected for the study as it considers both rock and soil site condition with interface faulting system. The attenuation relationships are appropriate for earthquakes of magnitude 5 and greater and for distance of 10 to 500km. As we have considered 300km x 300km rectangular area and magnitudes greater than 4 this attenuation relation for subduction zone will be appropriate for this thesis work.

2.8 RESEARCH GAP

Current study area lies in Province 1 of Nepal which has been unexplored yet regarding study of seismic hazard estimation. (Bhattarai G. , 2010), in his research work has done PSHA work for Biratnagar Sub-Metropolitan City, southern part of Province 1 of Nepal. But till date no any research work was done for Sankhuwasabha district.

3. METHODOLOGY

A detailed flow chart of the adopted methodology of this research work is presented in figure.



3.1 Seismic Hazard Assessment

Seismic hazard assessment involves the quantitative estimation of ground shaking hazards at a particular area. Seismic hazards can be analyzed deterministically as and when a particular earthquake scenario is assumed, or probabilistically, in which uncertainties in earthquake, size, location, and time of occurrence are explicitly considered (Kramer, 2007).

With PSHA, we consider all possible earthquake events and resulting ground motions, along with their associated probabilities of occurrence, in order to find the level of ground motion intensity exceeded with some tolerably low rate. At its most basic level, PSHA is composed of five steps.

1. Identify all earthquake sources capable of producing damaging ground motions.
2. Characterize the distribution of earthquake magnitudes (the rates at which earthquakes of various magnitudes are expected to occur).
3. Characterize the distribution of source-to-site distances associated with potential earthquakes.
4. Predict the resulting distribution of ground motion intensity as a function of earthquake magnitude, distance, etc.
5. Combine uncertainties in earthquake size, location and ground motion intensity, using a calculation known as the total probability theorem.

From the above steps seismic hazard curves for individual source zones can be obtained and combined to estimate the total hazard at a particular site. Let Ns, Nr and Nm be the number of sources in the region, different earthquake magnitudes and different source to site distances, the total exceedance rate (Kramer, 2007) for the region can be estimated by,

$$\lambda_{y^*} = \sum_{i=1}^{N_s} \sum_{j=1}^{Nr} \sum_{k=1}^{Nm} \nu_{iM_{min}} P\left[Y > y^* \middle| m_j, r_k\right] P\left[M = m_j\right] P[R = r_k] \tag{3.1}$$

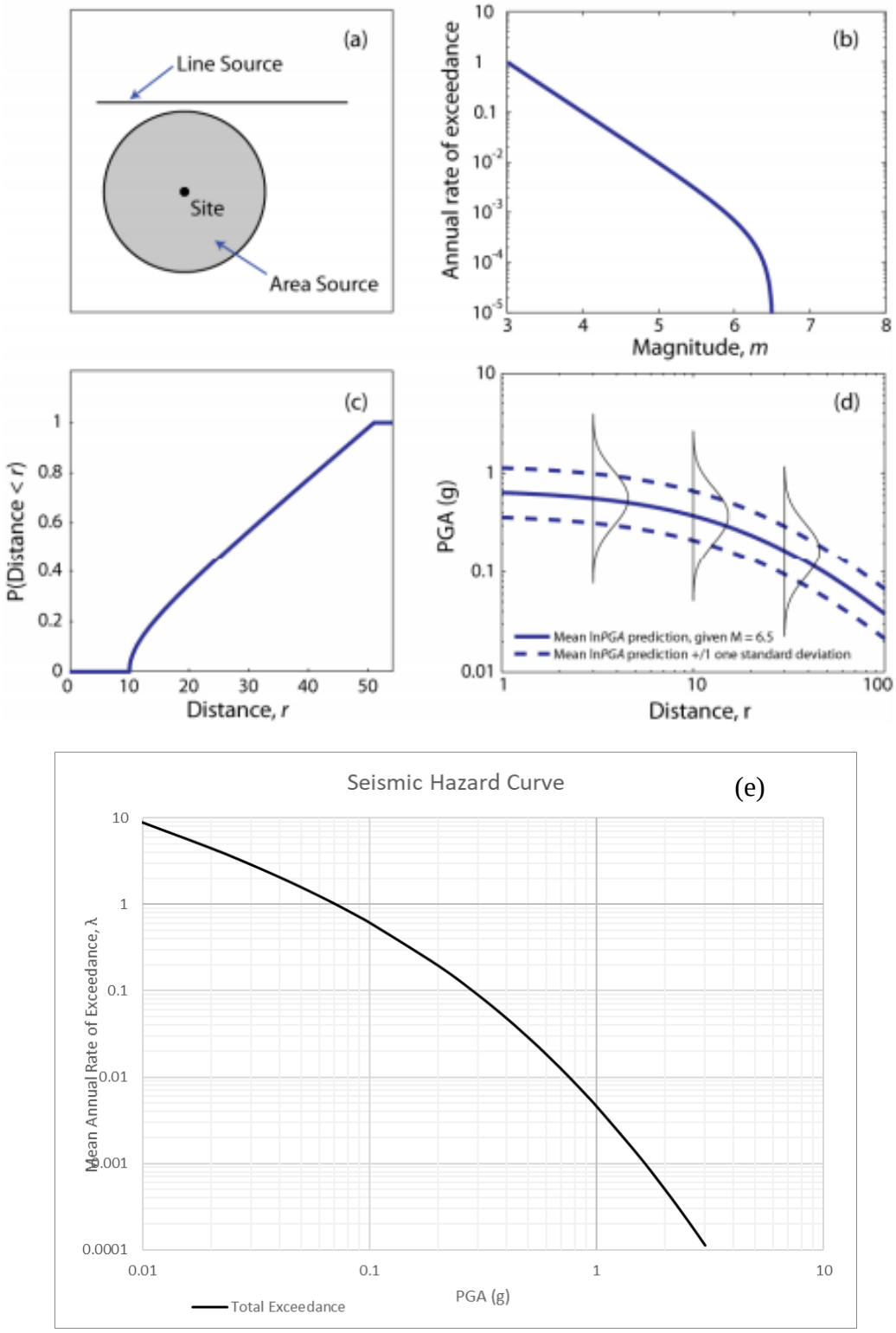


Figure 3.1.1 Schematic illustration of the basic five steps in probabilistic seismic hazard analysis. (a) Identify earthquake sources. (b) Characterize the distribution of earthquake magnitudes from each source. (c) Characterize the distribution of source-to-site distance from each source. (d) Predict the resulting distribution of ground motion intensity. (e) Combine information from parts a-d to compute the annual rate of exceeding a given ground motion intensity. (Author's computation)

3.1.1 Nature and source of data

The data used in the formation of earthquake catalogue was acquired from the following online earthquake database:

- United States Geological Survey (USGS)
- International Seismological Center (ISC)
- Incorporated Research Institutions for Seismology (IRIS)
- Nepal Seismological Center (NSC)

The above listed database sources provide instrumental earthquakes with its coordinates, magnitude, type of magnitude, depth and time etc.

Data taken from all these sources includes following main characteristics:

Table 3.1.1 Characteristics of earthquake data

SN.	Characterisitcs	Formats and Values
1	Coordinates	30.2252N 24.8027S 90.2686E 84.1997W
2	Magnitude Range	4≤Mw≤9
3	Magnitude Type	Ms, Mb, Ml, Mw
4	Depth	0 to 300km
5	Date	1255 to 2020
6	Output format	Text File

Following charts shows the numbers of events taken from various online database:

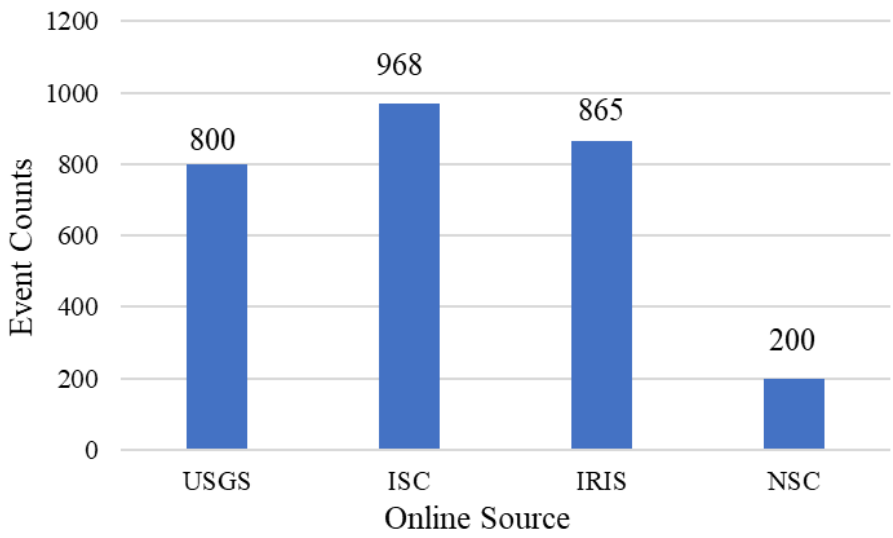


Figure 3.1.2 Sources and Earthquake Events

After removing duplicate events and declustering total numbers of earthquake events considered are 1124.

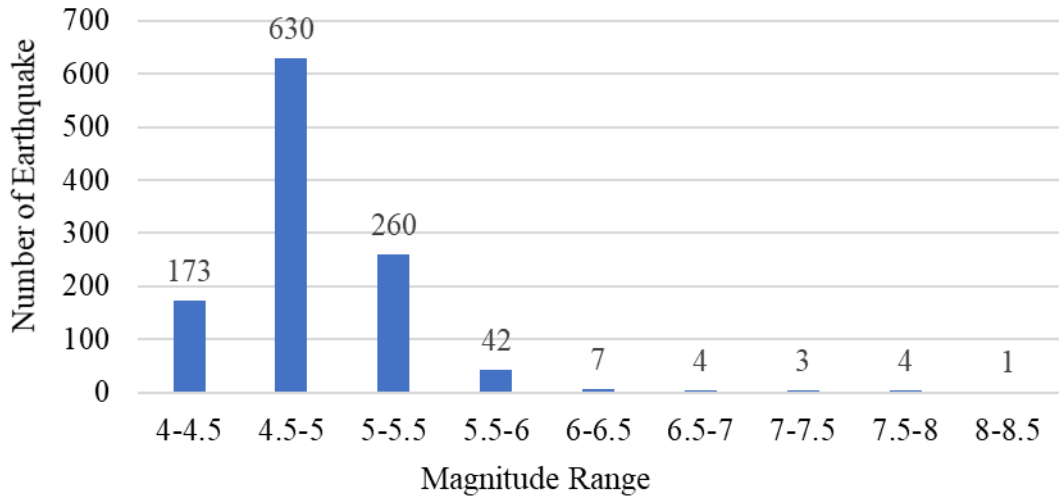


Figure 3.1.3 Earthquake event count by magnitude range

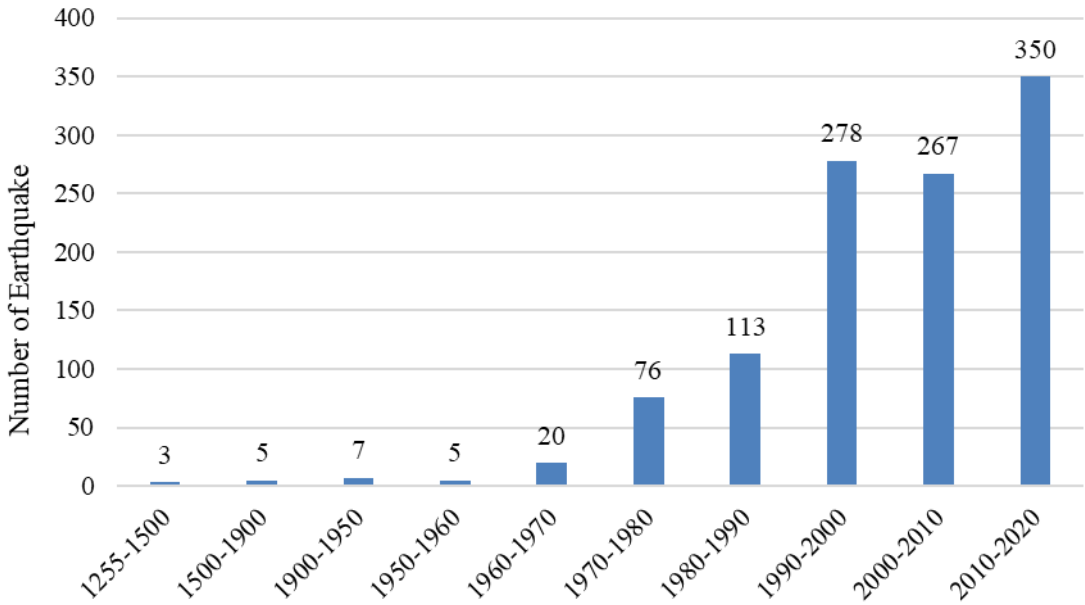


Figure 3.1.4 Number of earthquake (Mw>4) for different time intervals (after magnitude conversion)

3.1.2 Earthquake Catalogue

Sankhuwasabha district is in 27°06′N 86°57′E to 27°55′N 87°40′E and thus, all the earthquake data residing within the 600km x 600km rectangle area enclosed by 30°13′N 84°12′E to 24°48′N 90°16′E was collected. The earthquake catalogue was formed merging the data from U.S. Geological Survey (USGS, 2020), International Seismological sources (ISC, NEIC), Nepal Seismological center (NSC), Ambraseys and Douglas 2004.

The combined catalog merged from different sources has common attributes like Date Time, Longitude, Latitude, Magnitude and Depth.

Date Time	*Magnitude (Mw)*	Longitude	Latitude	Depth	Year	Magnitude
08/13/1924 23:57	5.82	90	29.5	35	1924	5.82
07/02/1930 21:03	7.1	90.18	25.93	15	1930	7.1
03/24/1932 16:08	5.82	90	25	35	1932	5.82
01/15/1934 8:43	8.18	86.59	26.88	15	1934	8.18
05/21/1935 4:22	6.27	89.25	28.75	140	1935	6.27
02/11/1936 4:48	5.82	87	27.5	50	1936	5.82
01/29/1938 4:13	5.76	87	27.5	35	1938	5.76

Figure 3.1.5 Attributes of Earthquake Event data

3.1.3 Earthquake Magnitude Conversion

Earthquake size can be described by quantitative and qualitative measures. Intensity is qualitative measure based on human reaction at any location during earthquake and structural and nonstructural damage in structures. For earthquakes that occurred before the development of modern seismic instruments, intensity is used to estimate the location and size of earthquakes.

Richter or Local magnitude scale, M_L, is defined as the logarithm (base 10) of the maximum trace amplitude (in micrometers) of local earthquakes recorded on typical Wood Anderson seismographs (T_0 = 0.8s, critical damping = 0.8, V =2800) located 100km from the epicenter of the earthquake. It is most common measure of the size of an earthquake.

Surface wave magnitude is based on the amplitude of Rayleigh waves with a period of about 20 seconds. Body wave magnitude is based on the amplitude of first few cycles of p-waves.

The composite earthquake catalogue contains earthquakes collected from different sources, which have reported different types of magnitudes (e.g., moment magnitude (Mw), body wave magnitude (Mb), surface wave magnitude (Ms) and local magnitude (ML)). Moment magnitude is generally considered the best overall estimate of an

earthquakes (Kanamori, 1977); (Hanks and Kanamori, 1979), therefore it is preferred in the study of seismic activity, seismotectonic and seismic hazard.

In order to develop a comprehensive catalog of consistent magnitude, the other types of magnitudes (Mb, Ms and M_l) were converted into moment magnitude (Mw) using the empirical relations (Equation (3.2), Equation (3.3) and Equation(3.4)) from (Scordilis, 2006).

$$Mw = 0.67(\pm 0.005) * Ms + 2.07\ (\pm 0.03) \tag{3.2}$$

(For shallow earthquakes (depth < 70km) and Ms between 3.0 and 6.1),

$$Mw = 0.99(\pm 0.02) * Ms + 0.08(\pm 0.13) \tag{3.3}$$

(For Ms between 6.2 and 8.2)

$$Mw = 0.85(\pm 0.04) * Mb + 1.03(\pm 0.23) \tag{3.4}$$

(For Mb between 3.5 and 6.2)

3.1.4 Declustering

An earthquake catalogue may contain fore-shocks and aftershocks in addition to mainshocks. A large size earthquake is generally followed by a large number of aftershocks. The foreshocks and aftershocks are dependent events to the mainshock. In the study of seismotectonic, seismic activity and probabilistic seismic hazard assessment, a catalogue of independent earthquake is required which follow the Poisson distribution.

So far, in most studies conducted to decluster earthquake catalogs, the (Reasenberg, 1985) linked-window method of the variant of (Gardner & Knopoff, 1974) window windowing method, a specified distance interval L(M) and a specified time interval T(M) have been defined for each earthquake with a magnitude of M. Subsequent earthquakes will be identified as an aftershock if they occur within this time-space window. Foreshocks will be identified the same as aftershocks. The largest shock in a sequence has the largest time-space window and is a main-shock.

For this study purpose linked-window method (Cluster Method) presented by (Reasenberg, 1985) is used, in which earthquakes are associated with clusters by defining spatial and temporal interaction zone around the earthquakes. ZMAP version 7.0 has been used in Matlab to decluster the earthquake catalog. ZMAP is a set of tools driven by a graphical user interface (GUI), designed to help seismologist analyze catalog data (Wiemer, 2001). The input parameters for Cluster Method are presented in Table below. In this study, the standard parameters of (Reasenberg, 1985) method are used.

Table 3.1.2 Input parameters for the Cluster Method presented by (Reasenberg, 1985)

Parameter	Standard	Min	Max
τ_{min}	1	0.5	2.5
τ_{max}	10	3	15
p1	0.95	0.9	0.99
x_{meff}	4	0	1
x_k	0.5	1.6	1.8
r_{fact}	10	5	20

τ_{min} and τ_{max} are the minimum and maximum look-ahead time of observing the next earthquake at a certain probability, p1. x_k is the increase of the lower cutoff magnitude during clusters: $x_{meff} = x_{meff} + x_k M$, where M is the magnitude of the largest event in the cluster, x_{meff} is the effective lower magnitude cutoff for catalog, r_{fact} is the number of crack radii surrounding each earthquake within new events considered to be part of the cluster.

3.1.5 Removal of duplicate events

Since various sources were referred during the formation of the catalogue, many events were reported more than once. Most of them were removed during the removal of aftershocks as they lied inside the window used in the process of declustering. Few duplicates retained were checked and removed manually.

3.1.6 Delineation of Seismic Source Zones and Source Models

Identification and delineation of regional seismic source zones is important in conducting a seismic hazard analysis. For zones of different seismicity, separate catalogs need to be used. Three types of source models that can be considered are linear seismic, smooth gridded model, and aerial sources.

Smooth gridded model is used in this study to develop recurrence relationships. In this method, study area is divided into grids and seismic activity rate is obtained for each cell considering center of the cell as a site. In the present study grid size was adopted as 200km x 200km, and the cutoff magnitude was selected as Mw 4.0. All the earthquake greater than a cutoff magnitude inside as specific radius of each site are included in catalogue. Catalogue is grouped into magnitude interval (say M 0.5) and recurrence rates are developed.

3.1.7 Completeness Analysis

Historical earthquake few and instrumental earthquake have been increased towards the end, those earthquake data are not uniformly distributed. Due to lack of earthquake data, short samples in which small magnitude events are completely reported may not represent long term seismicity of larger magnitude events. Whereas, longer year samples lead to serious underestimates of mean rate of occurrence of lower magnitude

earthquakes. It is necessary to obtain the interval of complete reporting in each intensity class over which that class is homogenous.

The completeness of the catalog was analyzed using the method suggested by (Stepp, 1972), which will give the time interval in which the magnitude range is homogenous. In this method the sample mean is inversely proportional to the number of observations in the sample. This test method relies on the statistical property of the Poisson distribution highlighting time intervals during which the recorded earthquake occurrence rate is uniform. Supposing that earthquake occurrences follow a Poisson distribution, the Stepp test evaluates the stability of the mean rate of occurrences (λ) of events which fall in a predefined intensity range in a series of time windows(T). If λ is constant, then the standard deviation (σ) varies as $1/\sqrt{T}$. On the contrary, if λ is not stable, σ deviates from the straight line of the $1/\sqrt{T}$ slope. The length of the time interval at which no deviation from the straight line occurs defines the completeness time interval for the given intensity range (Stepp, 1972). The interval is visually determined from the plots.

Let x1, x2, x3, Xn be the number of events per unit time interval; then the unbiased estimate of the mean per unit interval for this sample is given by

$$\lambda = \frac{1}{n}\sum_{i=1}^{n} Xi \tag{3.5}$$

Where n is the unit time interval and its variance is $\sigma^2 = \frac{\lambda}{n}$, When the time interval is taken as one year, the standard deviation of the above equation becomes $\sigma^2 = \frac{\sqrt{\lambda}}{\sqrt{T}}$; where T is the sample length. If this process is assumed to be stationary, it can be concluded that the standard deviation behaves as $\frac{1}{\sqrt{T}}$ in the subinterval of the complete years of the sample. Hence, during this period, the mean rate of occurrence in a magnitude class will be constant.

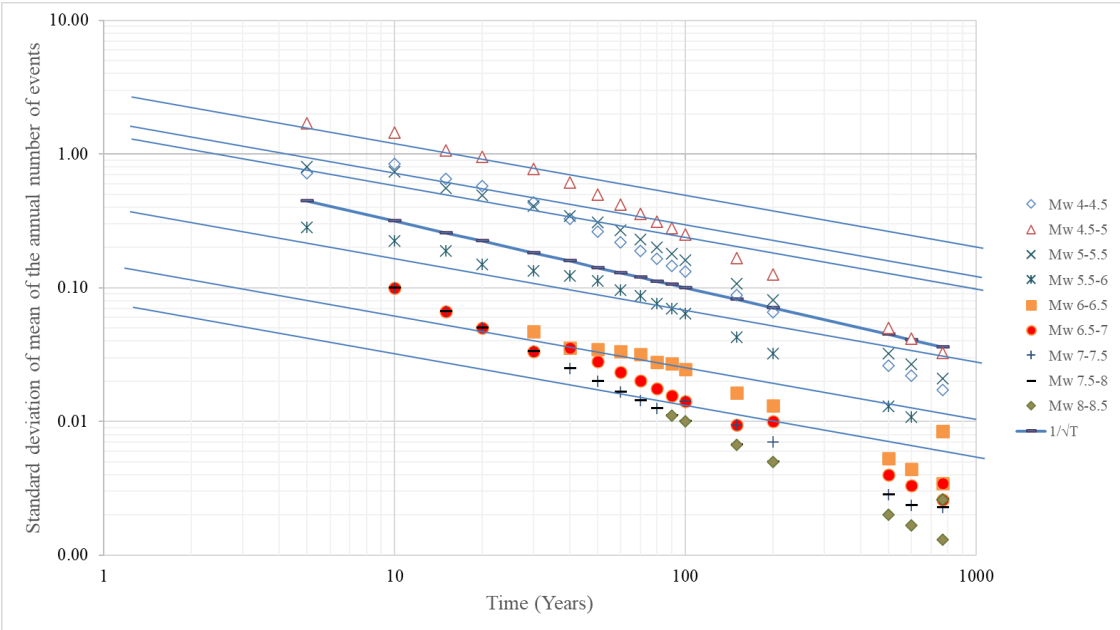


Figure 3.1.6 Completeness analysis using Stepp's method (Author's Calculation)

3.1.8 Evaluation of Seismicity Parameters

The distribution of earthquake size in a given period of time is expressed by Guttenberg Richter Recurrence Law (1997). The recurrence specifies average rate of exceedance of magnitude of earthquake for each source.

Gutenberg Richter law for earthquake recurrence is expressed as:

$$log\lambda_m = a - bm \tag{3.6}$$

Where λ_m is the mean annual rate of exceedance of magnitude M, 10^a is the mean yearly number of earthquakes of magnitude greater than zero, and b is a constant describing relative likelihood of large and small magnitude earthquakes.

3.1.9 Attenuation of ground motions

Seismic hazard estimation depends upon the attenuation relationship. An attenuation law is usually an empirical relationship that defines the transfer of ground motion from the source to a particular site as a function of magnitude of earthquake, source to site distance and geological characteristics of the site or tectonic environment faulting mechanism and medium of earthquake propagation.

Attenuation relationship is articulated as: Y = f (M, R, Pi), where Y is ground motion parameters of interest, M is the magnitude of earthquake, R is the source to site distance and Pi is the source path and local soil effect which may or may not be considered. Generally, with attenuation relationship, uncertainty associated with the expression i.e. standard deviation is also specified.

These attenuation relationships are based on the regional values, hence there are many types of attenuation relationships developed by researchers at different site condition.

There attenuation relationships can be categorized into four groups, shallow crustal earthquake in active regions (Campbell 1997, Campbell and Bozorgnia 2003, Abrahamson and Silva 1997, Boore et al. 1997), shallow crustal earthquakes in stable regions (Campbell 2003, Atkinson and Boore 1997), subduction zones(Molas and Yamazaki (1995), (Youngs et al., 1997), (Atkinson & Boore, 2008), (Zhao, et al., 2006), Extensional tectonic regimes (Spudich et al. 1997).

Most of the earthquakes occurring in Nepal are considered to be interface events due to subduction of Indian plate beneath the Eurasian plate. Hence, in this research work attenuation relationship suitable for subduction zone proposed by Youngs et al. (1997) is used.

For Rock Site,

$$\ln(y) = 0.2418 + 1.414M + C_1 + C_2(10 - M)^3 + C_3 \ln\left(R_{rup} + 1.7818e^{0.554M}\right) + 0.00607H + 0.384Z_T \quad (3.7)$$

Standard Deviation = $C_4 + C_5M$

Where, y is spectral acceleration in g, M is moment magnitude, Rrup is closest distance to rupture (km), H is depth (km) and Z_T coefficient for source type which is 0 for intraface event and 1 for interslab event. The coefficients C_1, C_2, C_3, C_4 and C_5 are given in the table.

Similarly, for the soil site the attenuation relationship is given by equation (7) and fig.

$$\ln(y) = -0.6687 + 1.438M + C_1 + C_2(10 - M)^3 + C_3 \ln\left(R_{rup} + 1.097e^{0.617M}\right) + 0.00648H + 0.364Z_T \quad (3.8)$$

Standard Deviation = $C_4 + C_5M$

Where, y, M, H and Z_T are the same as defined in equation (4.2.6). The coefficients C_1, C_2, C_3, C_4 and C_5 are given in the table.

Table 3.1.3 Coefficient for attenuation relations for rock site (after, Young's et al. 1997)

Period(s)	C_1	C_2	C_3	C_4*	C_5*
PGA	0.0	0.0	-2.552	1.45	-0.1
0.075	1.275	0.0	-2.707	1.45	-0.1
0.1	1.188	-0.0011	-2.655	1.45	-0.1
0.2	0.722	-0.0027	-2.528	1.45	-0.1
0.3	0.246	-0.0036	-2.454	1.45	-0.1
0.4	-0.115	-0.0043	-2.401	1.45	-0.1
0.5	-0.400	-0.0048	-2.360	1.45	-0.1
0.75	-1.149	-0.0057	-2.286	1.45	-0.1
1.0	-1.736	-0.0064	-2.234	1.45	-0.1
1.5	-2.634	-0.0073	-2.160	1.50	-0.1
2.0	-3.328	-0.0080	-2.107	1.55	-0.1
3.0	-4.511	-0.0089	-2.033	1.65	-0.1

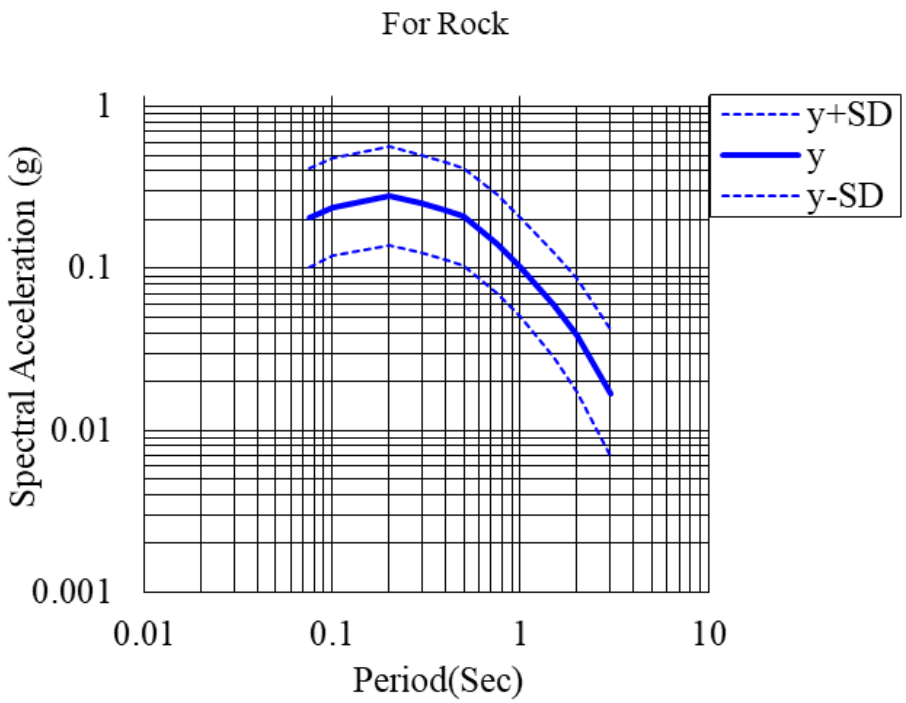


Figure 3.1.7 Spectral acceleration for rock site by attenuation relation for 5% damping in M = 7.5, Rrup = 50km and depth=15km interface earthquake (after, Young's et al. 1997)

Table 3.1.4 Coefficient for attenuation relations for rock site (after, Young's et al. 1997)

Period(s)	C_1	C_2	C_3	C_4*	C_5*
PGA	0.0	0.0	-2.329	1.45	-0.1
0.075	2.400	-0.0019	-2.697	1.45	-0.1
0.1	2.516	-0.0019	-2.697	1.45	-0.1
0.2	1.549	-0.0019	-2.464	1.45	-0.1
0.3	0.793	-0.0020	-2.327	1.45	-0.1
0.4	0.144	-0.0020	-2.230	1.45	-0.1
0.5	-0.438	-0.0035	-2.140	1.45	-0.1
0.75	-1.704	-0.0048	-1.952	1.45	-0.1
1.0	-2.870	-0.0066	-1.785	1.45	-0.1
1.5	-5.101	-0.0114	-1.470	1.50	-0.1
2.0	-6.433	-0.0164	-1.290	1.55	-0.1
3.0	-6.672	-0.0221	-1.347	1.65	-0.1
4.0	-7.618	-0.0235	-1.272	1.65	-0.1

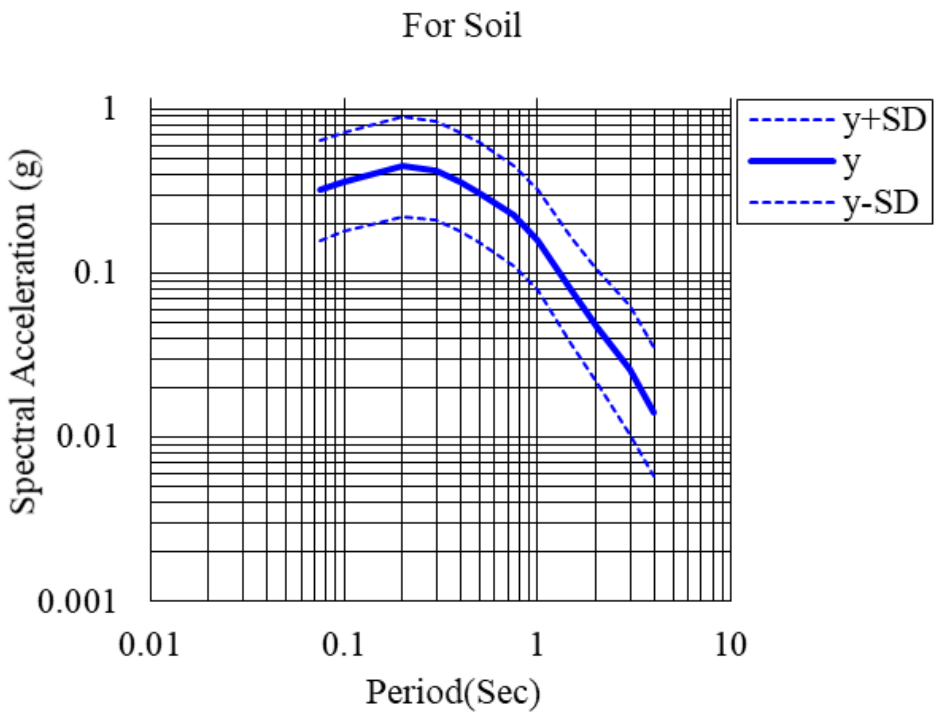


Figure 3.1.8 Spectral acceleration for soil site by attenuation relation for 5% damping in M=7.5, Rrup = 50km and depth = 15km interface earthquake (after, Young's et al. 1997)

3.1.10 Seismic Hazard Curve

The last step of PSHA is combining all uncertainties to obtain the probability that the ground motion parameter will be exceeded during the particular time period. (Kramer, 2007) gives basic theory mentioned for developing seismic hazard curves.

The seismic hazard curve for individual source zone is obtained at first and combined to get the hazard for the particular site. The probability that the particular value of ground motion parameter y* exceeds Y is calculated for one possible earthquake at one possible source location and then multiplied by the probability that the particular magnitude earthquake would occur at the particular location. This process is repeated for all possible magnitudes and locations with the probabilities of each summed.

The probability that a ground motion parameter Y will exceed a particular value y* can be computed by using total probability theorem. i.e.

$$P[Y > y^*] = P[Y > y^*|X]P[X] = \int P[Y > y^*|X]\, f_x\,(X)dx \tag{3.9}$$

Where, X is a vector of random variable that influences Y. In most cases the quantities in X are limited to the magnitude, M and distance, R, assuming that M and R are independent, the probability of exceedance can be written as

$$P[Y > y*] = \int\int P[Y > y^*|m,r]\;\; f_M(m)f_R(r)dm.dr \tag{3.10}$$

Where, $P[Y > y^*|m,r]$ is obtained from the predictive relationship and $f_M(m)$ and $f_R(r)$ are the probability density functions for the magnitude and direction, respectively.

If the site of interest is in a region of N_s potential earthquake sources, each of which has an average rate of threshold magnitude exceedance, $\nu_i = \exp\,(\alpha_i - \beta_i m_0)$, the total average exceedance rate for the region will be given by

$$\lambda_{y^*} = \sum_{i=1}^{N_s} \nu_{iM_{min}} \int\int P[Y > y^*|m,r]\;\; f_{Mi}(m)f_{Ri}(r)dm.dr \tag{3.11}$$

The next step is to divide the possible ranges of magnitude and distance into N_M and N_R segments, respectively the average exceedance rate can then be estimated by

$$\lambda_{y^*} = \sum_{i=1}^{N_s}\sum_{j=1}^{Nr}\sum_{k=1}^{Nm} \nu_{iM_{min}}\, P\big[Y > y^*\big|m_j, r_k\big]P\big[M = m_j\big]P[R = r_k] \tag{3.12}$$

The standard Gutenberg-Richter law predicts the non-zero mean rate of exceedance for magnitudes up to infinity. We are concerned with the earthquake greater than magnitude (M=4) since greater size earthquake generally produces maximum level of shaking. So bounded recurrence relation law is used to express the certain maximum magnitude M_{max} associated with each zone the value of which is greater than minimum magnitude M_{min}. The probability density function for Gutenberg Richter law with lower and upper bound magnitude is given by

$$f_M(m) = \frac{\beta \exp\left[-\beta(m - m_{min})\right]}{1 - \exp\left[-\beta(m_{max} - m_{min})\right]} \tag{3.12}$$

3.1.11 Uniform Hazard Spectra

Uniform hazard spectra of Equal hazard spectra represent the spectral values with the same mean recurrence interval developed on the basis of probabilistic basic. A uniform hazard spectrum is developed by first computing the curve for hazard of spectral periods using response spectral attenuation relations. The hazard is computed independently considering all sources independently for each spectral period. For a selected return period, the spectral values (spectral accelerations) for each period is measured from the hazard curves. The spectral values are then plotted at their respective spectral periods.

In order to determine the mean recurrence interval, different models can be adopted. Usually the probability model adopted for this work is Poisson's model. Poisson's model of probability distribution assumes no evidence of past events. That is, it assumes there is no any effect of past earthquake on the occurrence of present event. Poisson's model for probability distribution function is given by:

$$P(N \geq 1) = 1 - e^{-\lambda m.t} \tag{3.13}$$

Which means that probability of exceedance of at least 1 event having all ground motion parameters to be greater than a particular target ground motion parameter such that $Y > y^*$ within time t is given by function $1 - e^{-\lambda m.t}$, where, λ_m is mean annual rate of exceedance of that particular event. This model is used to find the spectral acceleration having uniform probability of exceedance. At exceedance probability of 10% in 50 years, by using equation (3.13), if P = 0.1 and t = 50 years, λ_m=0.0021 with return period of 1/0.0021 or 475 years. Similarly, for P=0.02 and t = 50 years, λ_m=0.0004 with return period of 2475 years.

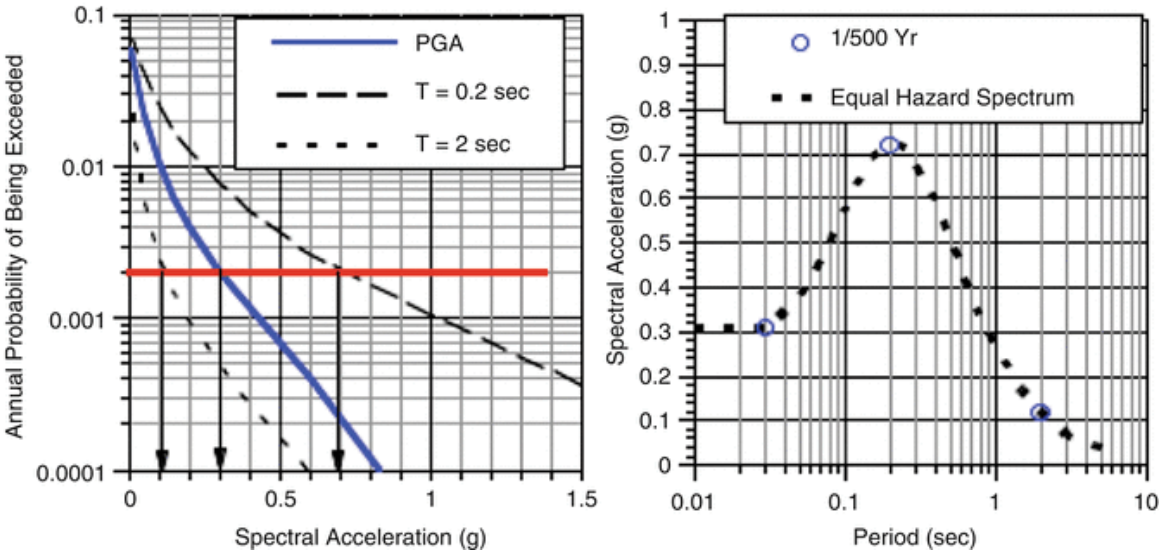


Figure 3.1.9 Procedure for developing uniform hazard spectra for a return period of 500 year (after Abrahamson, unpublished)

4. PRESENTATION AND ANALYSIS OF DATA

4.1 Source Characteristics

The area enclosed by 30°13′N 84°12′E to 24°48′N 90°16′E was divided into 200km x 200km grid size and seismicity within 600km x 600km was considered.

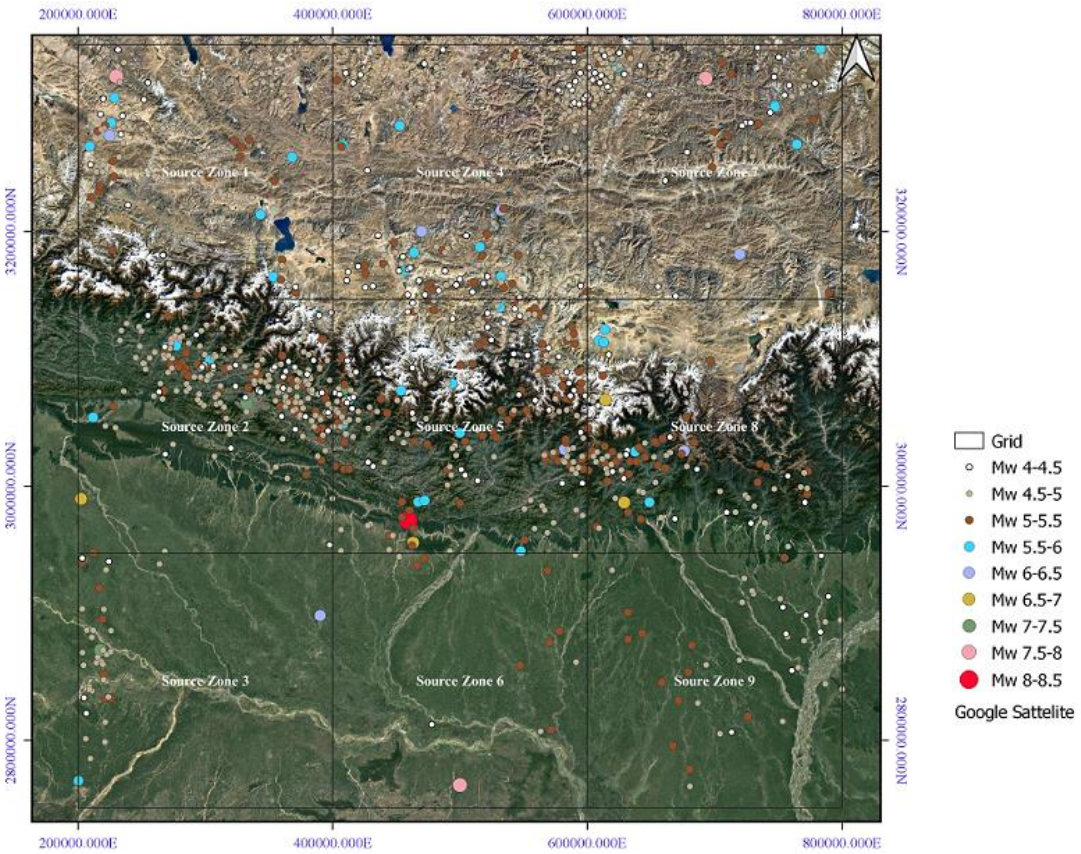


Figure 4.1.1 Delineation of total area into numbers of gridded sources showing earthquake events of different magnitude (Author's Calculation)

The grid coordinate of earthquake sources is shown Table 4.1.1 which represents the corner coordinates of each grid.

Table 4.1.1 Corner Coordinate of gridded Source Zones

	Top Left X	Top Left Y	Bottom Right X	Bottom Right Y
Source 1	200000	3347060	400000	3147060
Source 2	400000	3347060	600000	3147060
Source 3	600000	3347060	800000	3147060
Source 4	200000	3147060	400000	2947060
Source 5	400000	3147060	600000	2947060
Source 6	600000	3147060	800000	2947060
Source 7	200000	2947060	400000	2747060
Source 8	400000	2947060	600000	2747060
Source 9	600000	2947060	800000	2747060

4.2 Maximum Magnitudes

Maximum magnitude of individual source zone is taken as controlling earthquake within the region after declustering presented in table below:

Table 4.2.1 Maximum Magnitude of gridded sources

Sources	Max. Magnitude
Source1	7.66
Source2	6.2
Source3	7.7
Source4	7.9
Source5	8.18
Source6	6.9
Source7	7.1
Source8	7.8
Source9	5.45

4.3 Gutenberg and Richter Recurrence law Parameters

After characterizing the earthquake sources, logarithmic value of the rate of exceedance of earthquakes falling in the particular sources are plotted against the earthquake magnitude in order to find out the G-R parameters. The slope of the plotted curve represents the “b” value while the rate of earthquake exceeding 0 magnitudes represents the “a” value.

Source Zone 1

Table 4.3.1 Quantitative distribution of instrumental records within considered magnitude interval for Source Zone 1

M >	Nm	Log(λm)
4	60	0.301
4.5	49	0.213
5	18	-0.444
5.5	10	-1.000
6	2	-1.699
6.5	1	-2.884
7	1	-2.884
7.5	1	-2.884

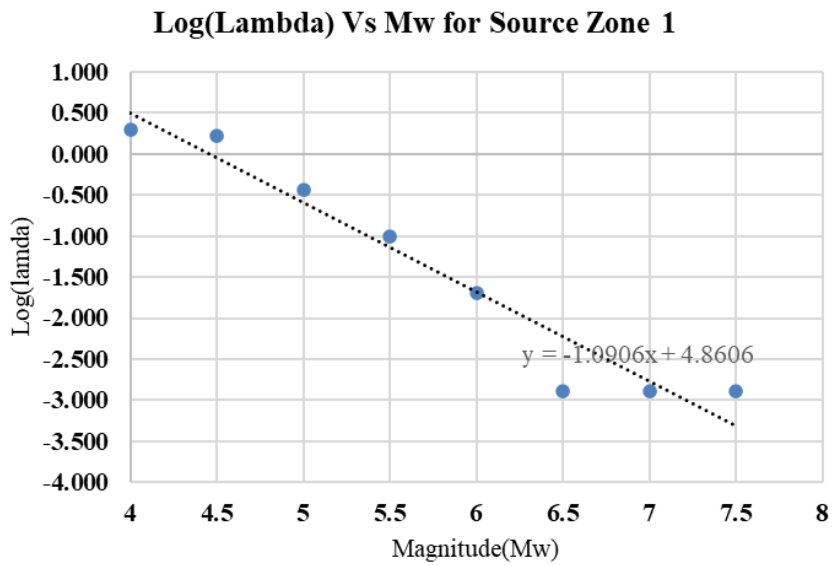


Figure 4.3.1 Gutenberg-Richter recurrence relationship curve for Source Zone 1

Source Zone 2

Table 4.3.2 Quantitative distribution of instrumental records within considered magnitude interval for Source Zone 2

M >	Nm	Log(λm)
4	182	0.783
4.5	140	0.669
5	30	-0.222
5.5	10	-1.000
6	2	-1.699

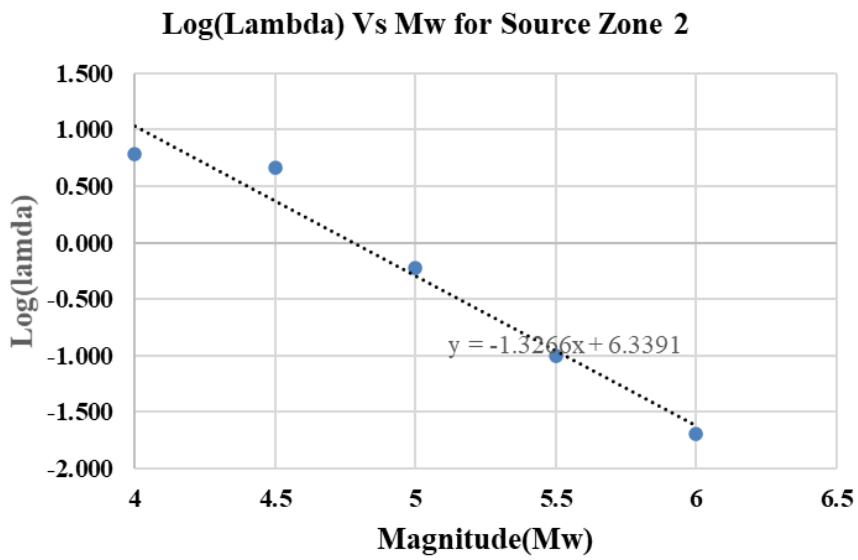


Figure 4.3.2 Gutenberg-Richter recurrence relationship curve for Source Zone 2

Source Zone 3

Table 4.3.3 Quantitative distribution of instrumental records within considered magnitude interval for Source Zone 3

M >	Nm	Log(λm)
4	147	0.690
4.5	120	0.602
5	33	-0.180
5.5	9	-1.046
6	2	-1.699
6.5	1	-2.884
7	1	-2.884
7.5	1	-2.884

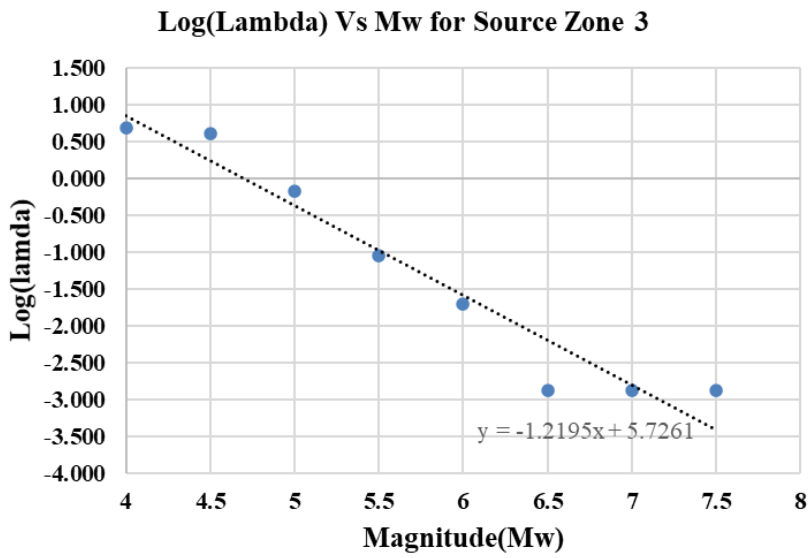


Figure 4.3.3 Gutenberg-Richter recurrence relationship curve for Source Zone 3

Source Zone 4

Table 4.3.4 Quantitative distribution of instrumental records within considered magnitude interval for Source Zone 4

M >	Nm	Log(λm)
4	191	0.804
4.5	161	0.730
5	36	-0.143
5.5	6	-1.222
6	3	-1.523
6.5	3	-2.407
7	2	-2.583
7.5	1	-2.884

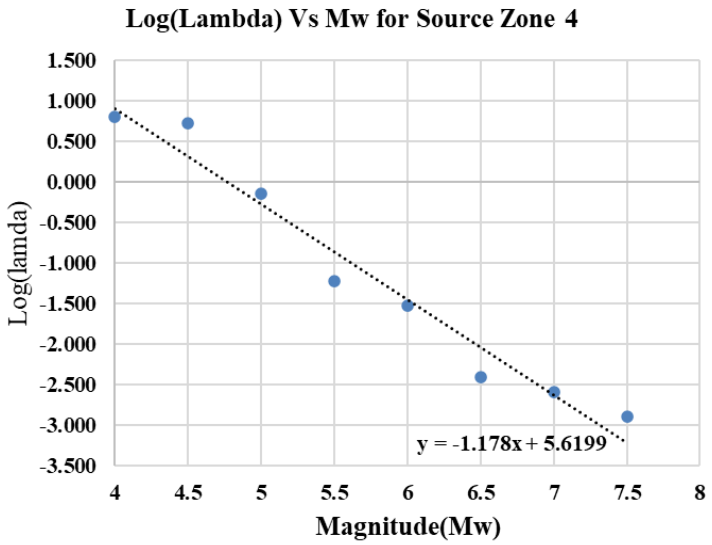


Figure 4.3.4 Gutenberg-Richter recurrence relationship curve for Source Zone 4

Source Zone 5

Table 4.3.5 Quantitative distribution of instrumental records within considered magnitude interval for Source Zone 5

M >	Nm	Log(λm)
4	212	0.849
4.5	174	0.763
5	55	0.041
5.5	14	-0.854
6	4	-1.398
6.5	3	-2.407
7	2	-2.583
7.5	1	-2.884
8	1	-2.884

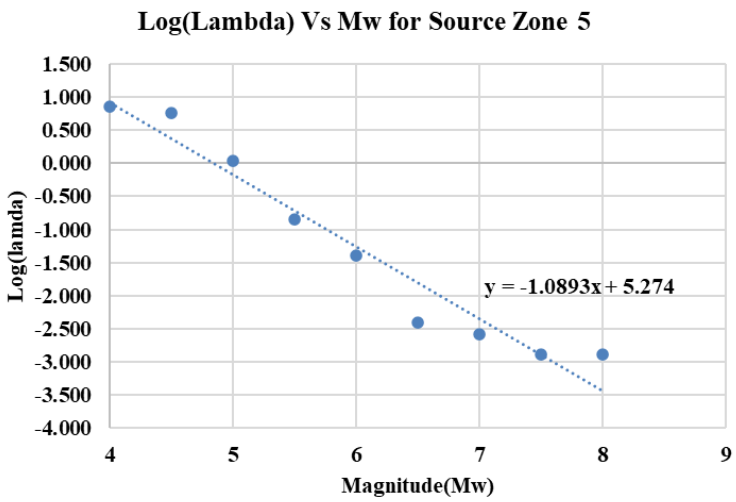


Figure 4.3.5 Gutenberg-Richter recurrence relationship curve for Source Zone 5

Source Zone 6

Table 4.3.6 Quantitative distribution of instrumental records within considered magnitude interval for Source Zone 6

M >	Nm	Log(λm)
4	65	0.336
4.5	52	0.239
5	27	-0.268
5.5	8	-1.097
6	3	-1.523
6.5	2	-2.583

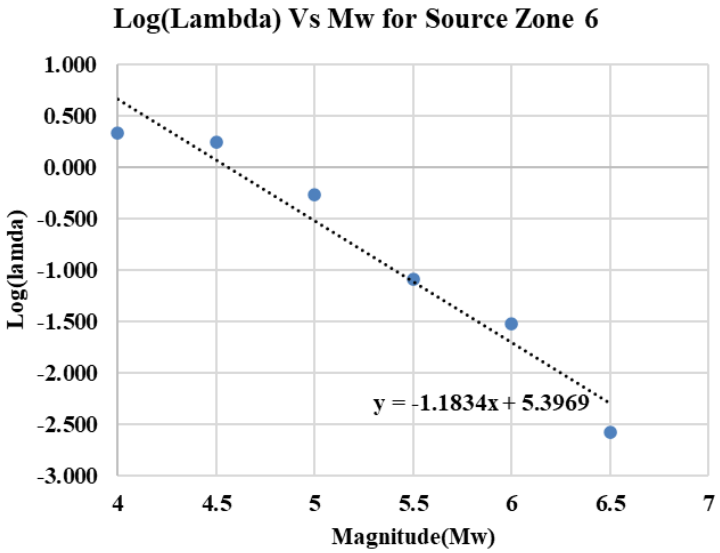


Figure 4.3.6 Gutenberg-Richter recurrence relationship curve for Source Zone 6

Source Zone 7

Table 4.3.7 Quantitative distribution of instrumental records within considered magnitude interval for Source Zone 7

M >	Nm	Log(λm)
4	33	0.041
4.5	29	-0.015
5	4	-1.097
5.5	1	-2.000
6	1	-2.000
6.5	1	-2.884
7	1	-2.884

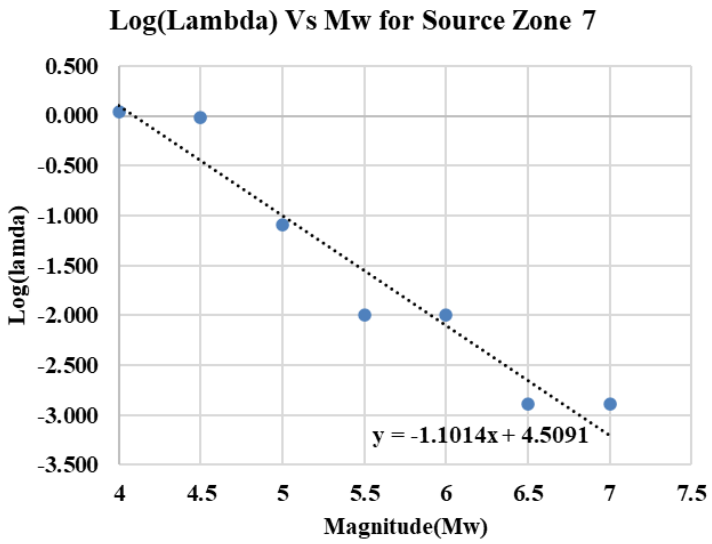


Figure 4.3.7 Gutenberg-Richter recurrence relationship curve for Source Zone 7

Source Zone 8

Table 4.3.8 Quantitative distribution of instrumental records within considered magnitude interval for Source Zone 8

M >	Nm	Log(λm)
4	7	-0.632
4.5	6	-0.699
5	5	-1.000
5.5	1	-2.000
6	1	-2.000
6.5	1	-2.884
7	1	-2.884
7.5	1	-2.884

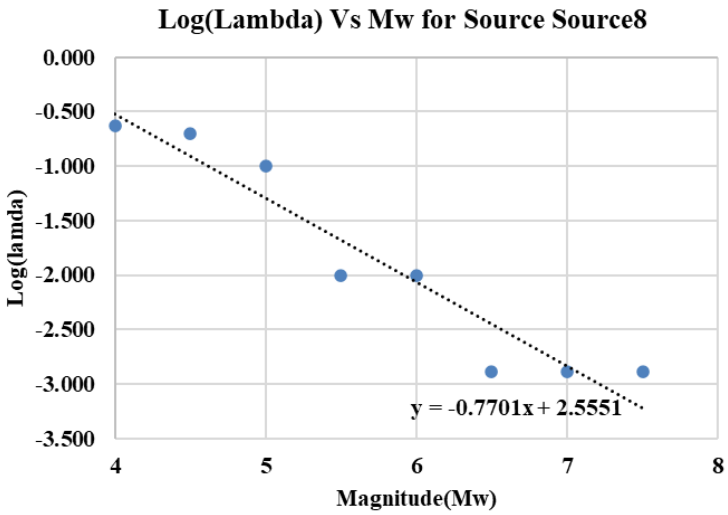


Figure 4.3.8 Gutenberg-Richter recurrence relationship curve for Source Zone 7

Source Zone 9

Table 4.3.9 Quantitative distribution of instrumental records within considered magnitude interval for Source Zone 9

M >	Nm	Log(λm)
4	29	-0.015
4.5	21	-0.155
5	4	-1.097

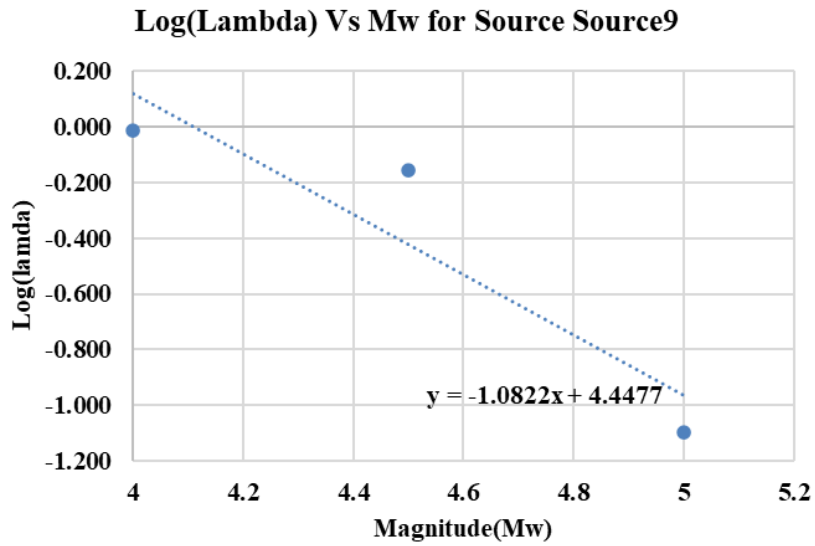


Figure 4.3.9 Gutenberg-Richter recurrence relationship curve for Source Zone 9

Considering minimum level of earthquake of moment magnitude 4 below which no damage occurs in the structures mean annual rate of exceedance of all the earthquakes above Mw=4 is calculated as:

$$\lambda_m \; or \; V_{Ai} = 10^{(a-bm)} \tag{4.1}$$

Using above equation (4.1) the mean annual rate of exceedance of earthquake of size 4 at all the sources are tabulated below:

Source Zones	G-R Relation	a	b	Mean Annual Rate of Exceedance (Vai or λm) = 10^(a-bm)
Source1	y = -1.0906x + 4.8606	4.861	1.091	3.149
Source2	y = -1.3266x + 6.3391	6.339	1.327	10.782
Source3	y = -1.2195x + 5.7261	5.726	1.220	7.049
Source4	y = -1.178x + 5.6199	5.620	1.178	8.089
Source5	y = -1.0893x + 5.274	5.274	1.089	8.257
Source6	y = -1.1834x + 5.3969	5.397	1.183	4.606
Source7	y = -1.1014x + 4.5091	4.509	1.101	1.269
Source8	y = -0.7701x + 2.5551	2.555	0.770	0.298
Source9	y = -1.0822x + 4.4477	4.448	1.082	1.315

4.4 Computation of Hazard

4.4.1 Magnitude Probability

Within a source zone maximum and minimum magnitude, the magnitude is divided into 10 ranges and probability of each magnitude is calculated using Gutenberg-Richter Law of upper and lower bound as represented by equation (4.2). Probability of magnitude occurring in a source is:

$$P(m) = P(mi < m < m_u) = f_M(m) * \Delta_M \tag{4.2}$$

Where, $f_M(m)$ is probability density function as defined in equation (3.12) and Δ_M is the magnitude interval between two consecutive magnitude in range.

Sample calculation of magnitude probability for Source Zone 1 is presented below and for other sources same process is used.

Sample calculation (Source zone 1)

Maximum Magnitude (M_{max}) = 7.66

Minimum Magnitude (($M_{\min}$) = 4

Dividing the magnitude range into 10 equal intervals, interval difference (Δ_M) = (7.66-4)/10 = 0.366

Using equation pdf, the probability of magnitude is calculated.

The probability of each magnitude in the range is shown in Table 4.4.1 and figure 4.4.1:

Table 4.4.1 Magnitude Probability

Magnitude, Mw	***Mean of Mw***	***Probability of Occurrence of magnitude P(M=m)***
4		
4.366	4.183	0.581
4.732	4.549	0.232
5.098	4.915	0.092
5.464	5.281	0.037
5.83	5.647	0.015
6.196	6.013	0.006
6.562	6.379	0.002
6.928	6.745	0.001
7.294	7.111	0.000
7.66	7.477	0.000

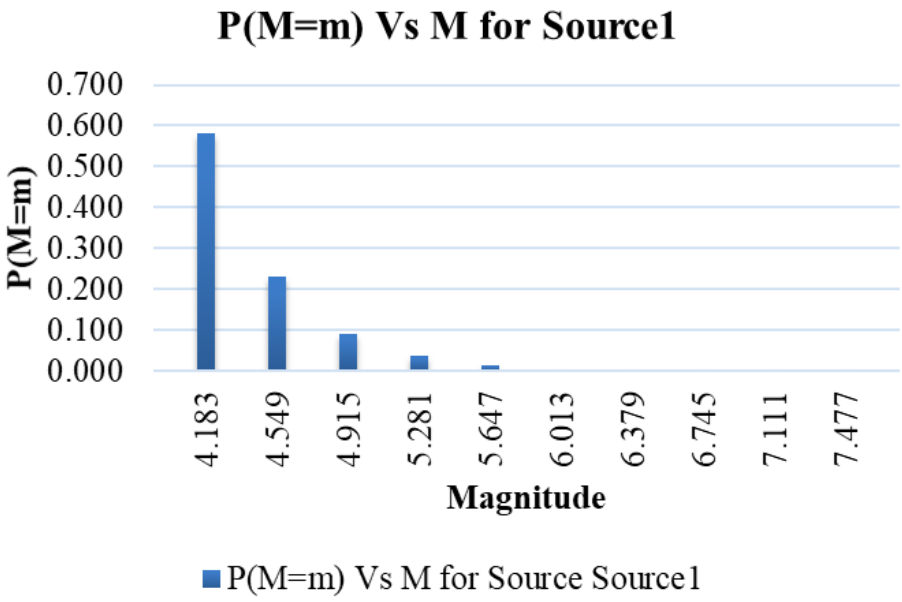


Figure 4.4.1 Magnitude Probability for Source Zone 1

4.4.2 Distance Probability

Each source zone is divided into 1000 x 1000 numbers of grids. The distance between the site to center of each grid is measured. Computation of source to site distance without a computer program is very tedious and time consuming, therefore a programming code in "Visual Basic Application (VBA)" is written for this work. The input parameters are corner coordinates of the source zone as well as site coordinate and the output are obtained in the form of distances of grid center from the site.

Once the source to site distances for all grid centers are obtained, the maximum and minimum distances are identified. Finally, uniform probability of each distance is calculated in the form of relative frequency of each distance range**.**

Sample Calculation of Source Zone 1

Site Coordinate: X = 489789.44, Y = 3096889.24

Source Coordinate:

Minimum distance of grid in Source Zone 1: 103 km

Maximum distance of grid in Source Zone 1: 383 km

For 10 divisions of distance range, interval of ranges $\Delta_{diff=}$(383-103)/10 = 28 km

After dividing the source zone into 1000x1000 numbers of grids, a total of 1000000 grids are obtained.

Total number of sources to site distance = 1000000

Thus, the distance from source i.e. grids to site is counted and number of each distance withing the range is used to find distance probability.

Table 4.4.2 Distance probability, Source Zone 1

Distance, R(km) (i)	*Average Distance. Rmid(km) (ii)*	*Number of distances in range (iii)*	*P(R=r) = [(iii)/1000000] (iv)*
103			
131	117	20326	0.020
159	145	55711	0.056
187	173	88475	0.088
215	201	120336	0.120
243	229	151807	0.152
271	257	182005	0.182
299	285	177357	0.177
327	313	119345	0.119
355	341	64459	0.064
383	369	20178	0.020

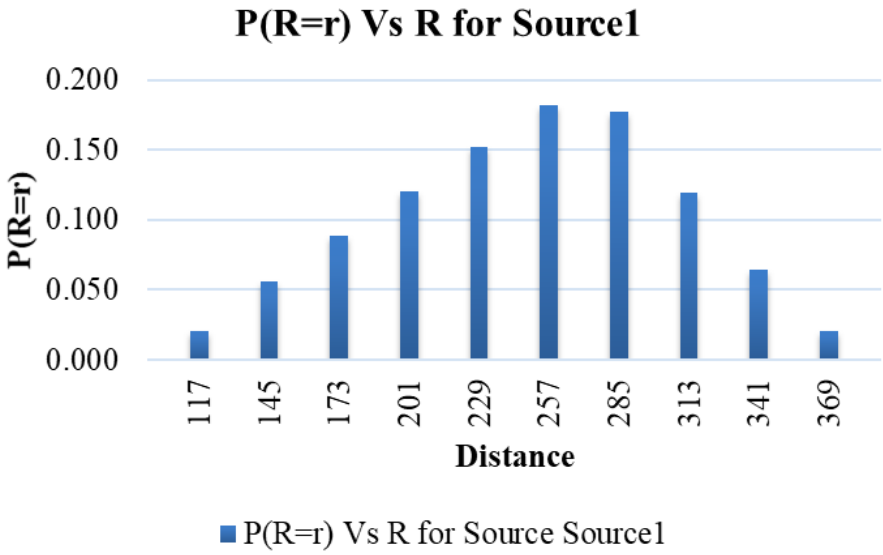


Figure 4.4.2 Distribution of uniform probability for Source Zone 1 to site distance

4.4.3 Probability of ground motion parameters

To find the spectral acceleration for a particular time period the attenuation relationship given by Young's et al. (1997) is used. The spectral acceleration calculated by attenuation relations represents the mean value in standard log normal distribution i.e. log of random variables in normal distribution. The probability of exceedance of target acceleration to this mean value is computed from the normal distribution table.

Table 4.4.3 Response spectral acceleration for 5% damping, Sa or y in g (for given magnitude and distance) for T=0sec due to Source A1.

T=0s		D(km)									
Ln(y)	D	117	145	173	201	229	257	285	313	341	369
		Hypocentral Distance (Rrup,km)									
M	R	117.427	145.344	173.289	201.249	229.218	257.194	285.175	313.160	341.147	369.135
4.183		-6.311	-6.789	-7.192	-7.540	-7.846	-8.119	-8.366	-8.592	-8.799	-8.990
4.549		-5.869	-6.334	-6.728	-7.069	-7.370	-7.639	-7.883	-8.105	-8.310	-8.499
4.915		-5.440	-5.891	-6.274	-6.608	-6.902	-7.167	-7.406	-7.625	-7.827	-8.014
5.281		-5.028	-5.462	-5.833	-6.157	-6.445	-6.703	-6.938	-7.153	-7.351	-7.535
5.647		-4.634	-5.049	-5.406	-5.720	-5.999	-6.251	-6.480	-6.690	-6.884	-7.065
6.013		-4.260	-4.654	-4.996	-5.297	-5.567	-5.810	-6.033	-6.238	-6.427	-6.604
6.379		-3.908	-4.279	-4.603	-4.891	-5.149	-5.384	-5.599	-5.798	-5.982	-6.153
6.745		-3.580	-3.926	-4.231	-4.503	-4.750	-4.974	-5.181	-5.372	-5.550	-5.716
7.111		-3.276	-3.596	-3.881	-4.137	-4.370	-4.583	-4.780	-4.963	-5.133	-5.293
7.477		-2.998	-3.291	-3.554	-3.793	-4.011	-4.212	-4.398	-4.572	-4.734	-4.887

D in Table 4.4.3 is epicentral distance. The bed rock level is assumed to be 15km below surface for the site, hence hypo central distance is calculated accordingly. Probability of exceedance of spectral acceleration of value 0.01g in log normal distribution of variables at this source is tabulated as below.

Table 4.4.4 Probability of exceedance of acceleration of 0.01g at Source Zone 1

P(M)	P(R)	0.020326	0.055711	0.088475	0.120336	0.151807	0.182005	0.177357	0.119345	0.064459	0.020178	Total Probability [$\lambda_{0.01g}$ * P(Mw) * P(R) * P(PGA>0.01g)]
0.58059		0.001827	0.0017474	0.0009852	0.00049	0.0002336	0.0001096	4.32E-05	1.22E-05	2.84E-06	3.94E-07	0.005451261
0.23155		0.001514	0.0016727	0.0010623	0.000583	0.0003024	0.0001524	6.39E-05	1.9E-05	4.63E-06	6.7E-07	0.00537474
0.09234		0.001133	0.0014559	0.0010499	0.000642	0.0003652	0.0001992	8.96E-05	2.82E-05	7.27E-06	1.1E-06	0.004972015
0.03683		0.000762	0.0011389	0.000938	0.000644	0.0004048	0.0002411	0.000117	3.95E-05	1.08E-05	1.73E-06	0.004297425
0.01469		0.000458	0.0007936	0.0007478	0.000579	0.0004051	0.0002655	0.00014	5.11E-05	1.5E-05	2.55E-06	0.003457563
0.00586		0.000247	0.0004901	0.0005267	0.000461	0.0003603	0.0002612	0.000151	5.99E-05	1.89E-05	3.45E-06	0.002579149
0.00234		0.00012	0.0002688	0.0003261	0.000321	0.0002807	0.0002259	0.000144	6.23E-05	2.14E-05	4.2E-06	0.001774715
0.00093		5.41E-05	0.0001323	0.0001779	0.000195	0.0001898	0.0001693	0.000119	5.65E-05	2.11E-05	4.5E-06	0.001119447
0.00037		2.29E-05	5.958E-05	8.661E-05	0.000104	0.000111	0.000109	8.44E-05	4.39E-05	1.79E-05	4.15E-06	0.000643225
0.00015		9.38E-06	2.52E-05	3.851E-05	4.92E-05	5.677E-05	6.05E-05	5.1E-05	2.89E-05	1.28E-05	3.24E-06	0.000335518
											Sum	0.030005058

Table 4.4.5 Mean Annual Rate of Exceedance of different level of spectral acceleration at rock site due to Source Zone 1, at T=0set and 5% damping

Sa/g	0.01	0.05	0.1	0.2	0.3	0.4	0.5	0.6	0.7	1	1.5	2	2.5
Rate of Exceedance	0.003	0.0006	7E-05	4.7E-06	7.7E-07	1.9E-07	5.8E-08	2.1E-08	8.7E-09	1E-09	6.7E-11	9E-12	2E-12

Finally, the mean annual rate of exceedance of different level of spectral acceleration at rock site due to all sources occurring at T=0 second and 5% damping are calculated and hazard curve is plotted. Zero period acceleration is also known as peak ground acceleration (PGA).

Mean annual rate of exceedance of different level of spectral accelerations at rock site due to all sources are calculated and tabulated in Table 6.5.4. The plot of hazard curve

is shown in Figure (4.5.1). The process is repeated for all sources for all the available time periods and finally the rate of exceedance of different spectral accelerations occurring at a particular time periods by all sources are summed up to get the total hazard at the given time period. The process is adopted for soil site and the hazard curve for both rock and soil site are plotted in Figure 4.4.5.

Table 4.4.6 Mean annual rate of exceedance of different level of spectral accelerations at rock site due to all sources zone 1, at T=0 sec and 5% damping.

PGA(g)	Source1	Source2	Source3	Source4	Source5	Source6	Source7	Source8	Source9	Total Exceedance
0.01	0.03001	0.67975	0.03504	0.18363	2.58904	0.06019	0.00214	0.00466	0.00073	3.58519878
0.05	0.00063	0.03628	0.00048	0.00506	0.58373	0.00109	0.00214	0.00466	1.8E-06	0.63408342
0.1	6.9E-05	0.00618	4.3E-05	0.00063	0.23658	0.00011	0.00214	0.00466	6.6E-08	0.25042096
0.2	4.7E-06	0.00073	2.4E-06	5.2E-05	0.07735	6.6E-06	0.00214	0.00466	1.5E-09	0.08495064
0.3	7.7E-07	0.00017	3.4E-07	9.5E-06	0.03557	1E-06	0.00214	0.00466	1.3E-10	0.04256542
0.4	1.9E-07	5.7E-05	7.5E-08	2.5E-06	0.01924	2.4E-07	0.00214	0.00466	2.1E-11	0.02610844
0.5	5.8E-08	2.3E-05	2.1E-08	8.5E-07	0.01148	7.2E-08	0.00214	0.00466	4.9E-12	0.01831625
0.6	2.1E-08	1E-05	7.4E-09	3.3E-07	0.00733	2.6E-08	0.00214	0.00466	1.4E-12	0.01415265
0.7	8.7E-09	5.2E-06	2.9E-09	1.4E-07	0.00492	1.1E-08	0.00214	0.00466	4.8E-13	0.01173555
1	9.9E-10	9.6E-07	2.9E-10	1.9E-08	0.00182	1.2E-09	0.00214	0.00466	3.6E-14	0.00863056
1.5	6.7E-11	1.2E-07	1.7E-11	1.5E-09	0.00052	8.3E-11	0.00214	0.00466	1.5E-15	0.00732683
2	8.7E-12	2.5E-08	2E-12	2.2E-10	0.0002	1.1E-11	0.00214	0.00466	1.2E-16	0.00700364
2.5	1.6E-12	7.1E-09	3.5E-13	4.5E-11	8.7E-05	2.1E-12	0.00214	0.00466	1.2E-17	0.00689547
3	4.1E-13	2.4E-09	8E-14	1.2E-11	4.4E-05	5.3E-13	0.00214	0.00466	8.5E-19	0.00685194

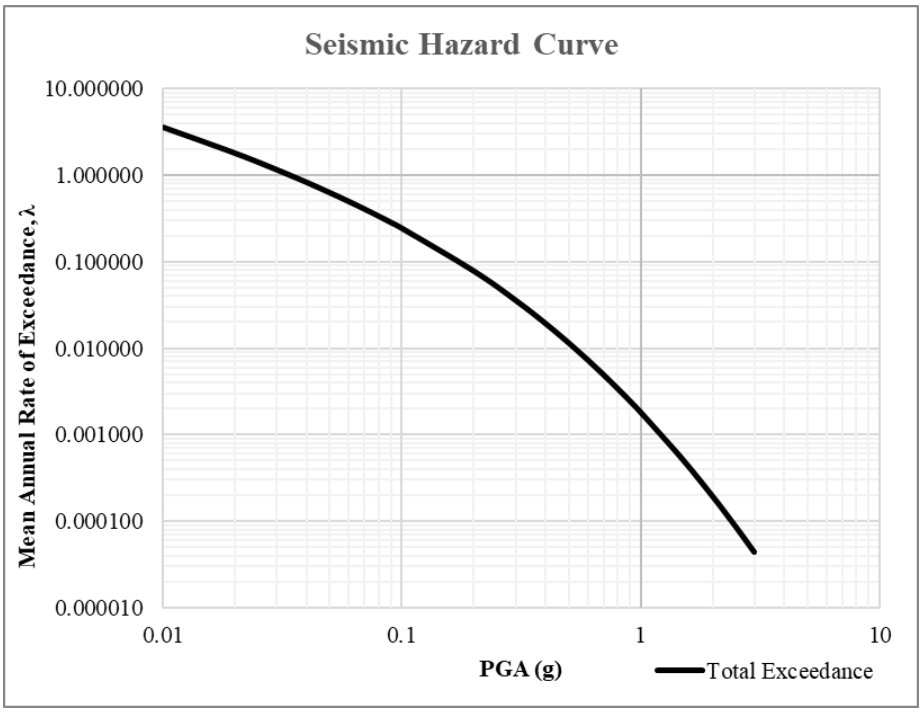


Figure 4.4.3 Seismic Hazard curve of PGA at rock site at 5% damping, due to all sources (Author's Calculation)

4.4.4 Computation of uniform hazard curves

Uniform hazard spectra represent the spectral acceleration having equal probability. The hazard curves for all spectral periods are plotted in the same graph. For uniform probability of 2% in 50 years (Collapse Prevention Level) and 10% in 50 years (Life Safety Level), the annual rate of exceedance of spectral acceleration becomes to be 0.002 and 0.0004 respectively i.e. with a return period of nearly 500 years and 2500 years. At this level of probability of exceedance, the spectral accelerations are obtained from hazard curves.

The total hazard for different spectral accelerations for different time periods of structure for rock and soil sites are shown in Table (4.4.5). This is used to find out the spectral acceleration values for exceedance rate of 0.002 and 0.004. The uniform hazard spectra are plotted in Figure (4.4.6) and (4.4.7).

The spectral accelerations are computed for the structures having 5% damping, as it is a representative value of viscous damping for the most of the structures. These spectral accelerations are plotted against respective time period to construct the uniform hazard curves for the site.

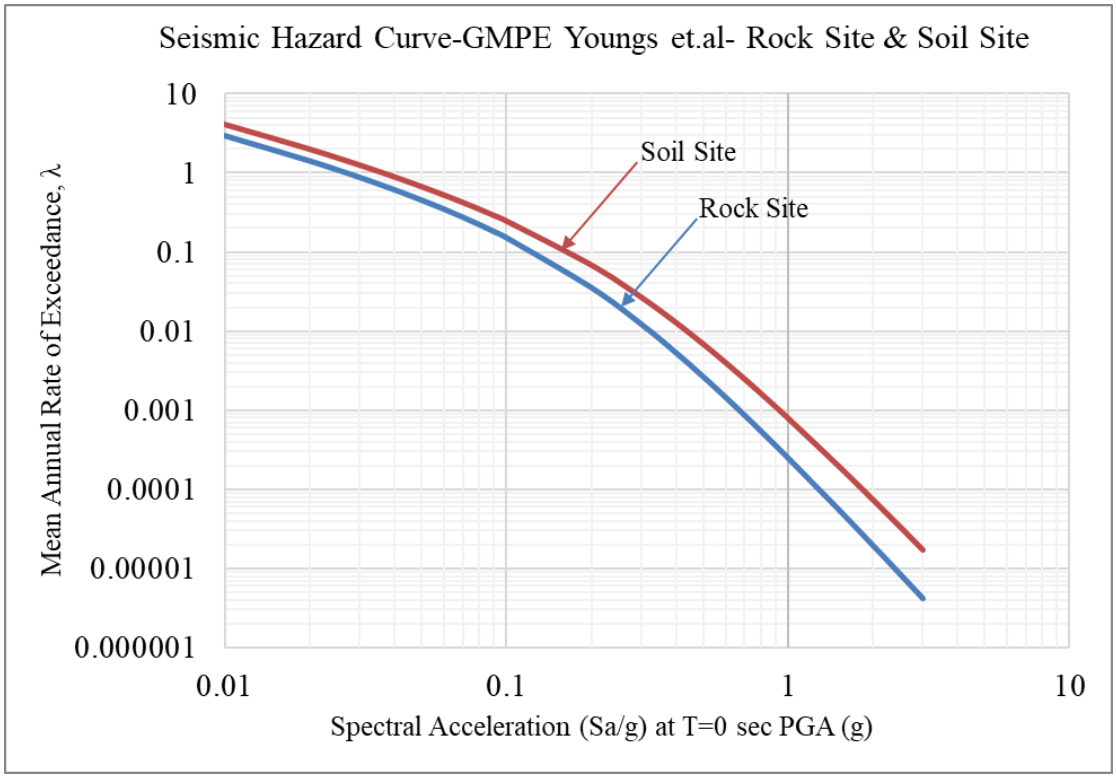


Figure 4.4.4 Comparison of Hazard curves for Rock and Soil Site

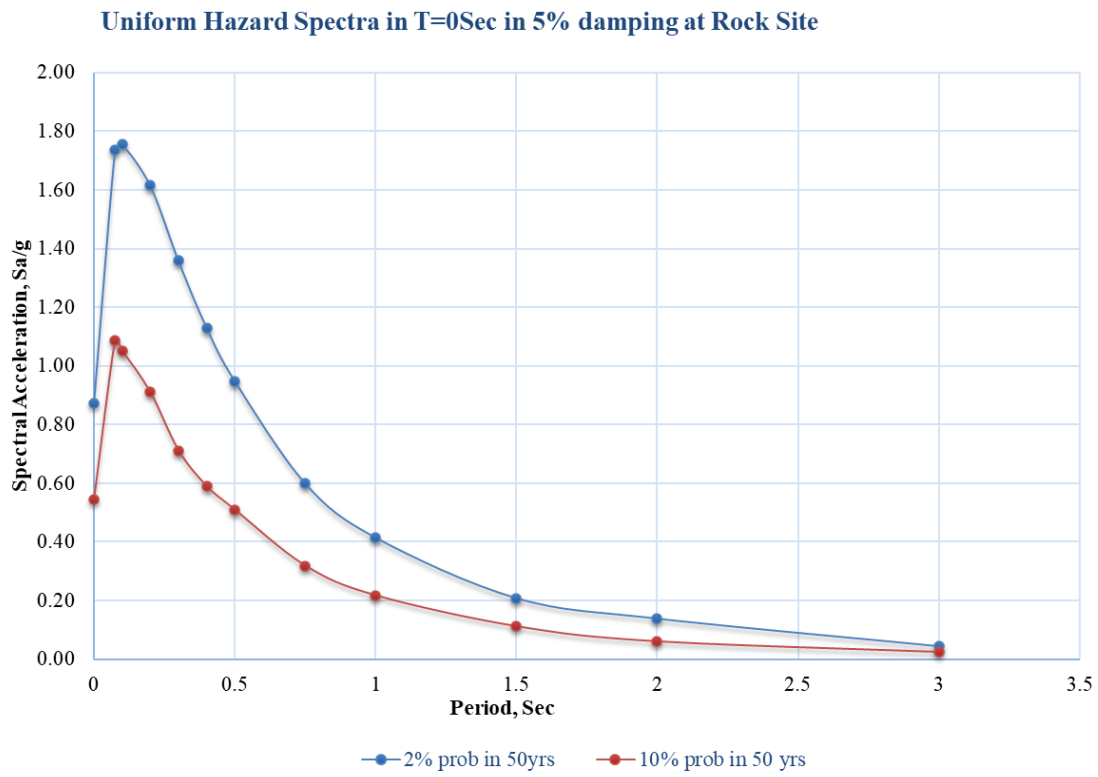


Figure 4.4.5 Uniform hazard spectra in 5% damping at rock site

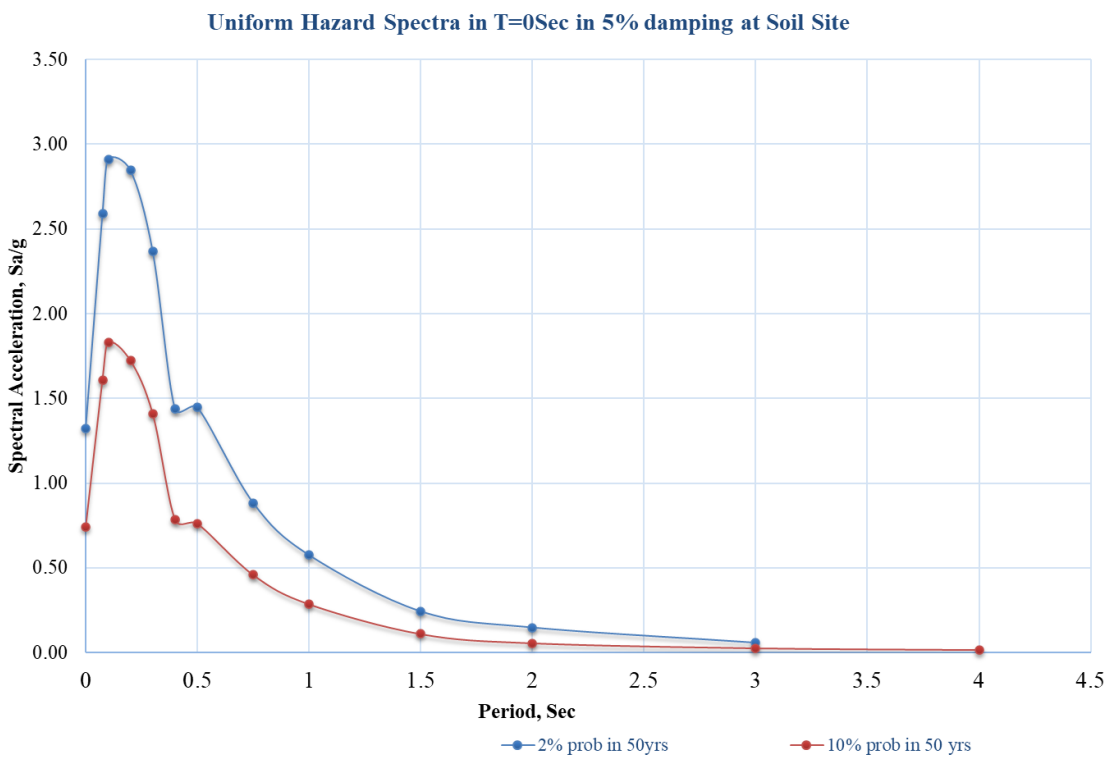


Figure 4.4.6 Uniform hazard spectra in 5% damping at soil site

5. CONCLUSION AND RECOMMENDATION

Probabilistic seismic hazard analysis for Sankhuwasabha district has been carried out to access peak ground acceleration and spectral acceleration for return periods 475, 2475 years using most recent earthquake information for the study area.

5.1 MAJOR FINDINGS

Total area has been delineated into 9 gridded source zones.

1. PSHA software module was developed with a functionality of downloading earthquake data from various International Federation of Digital Seismograph Network (FDSN) webservices such as USGS, ISC, IRIS etc. It removes duplicate earthquake data and incorporate output of ZMAP declustered events thereby dividing total study area into required numbers of gridded source zone and developing Gutenberg & Richter relation. Module currently supports Youngs et al.(1997) ,Gregor et al. (2002) and (Lin & Lee, 2008) ground motion attenuation relationships. Total Hazard curve and Uniform Hazard Spectra 2% probability in 50yrs and 10% probability in 50 yrs can generated.
2. A comprehensive and composite catalogue has been prepared for Sankhuwasabha district and its adjoining region between 30°13′N 84°12′E to 24°48′N 90°16′E. The developed catalogue contains 1124 events with Mw between 4.0 and 8.5, for the period between 1255 AD and 2020 AD and the magnitude of completeness is Mw 4.0.
3. The composite earthquake catalogue contains earthquake collected from different sources, which have reported different magnitude (e.g. moment magnitude (Mw), body wave magnitude (Mb), surface wave magnitude (Ms) and local magnitude (Ml). In order to develop as comprehensive catalog of consistent magnitude the all other types of magnitude (Mb, Ms and Ml) were converted into moment magnitude (Mw) using the empirical relations Equation (3.2), Equation (3.3) and Equation(3.4) from (Scordilis, 2006). The number of Mw>4 earthquake numbers is increased after magnitude conversion.
4. The total area of study i.e. 600km x 600km is divided into 200km x 200 km gridded zone. The selection and sizing of the grid zone directly affects the output of PSHA work.
5. A completeness test of the catalogue was performed, and larger earthquakes (Mw>7.0) found complete for entire span of time (since 1255 AD). The completeness test of the catalogue revealed that Mw 4, 4.5, 5, 5.5, 6, 6.5, and larger than Mw 7 are complete for 30,30,50,100,100,765.
6. Magnitude frequency relationship has been developed for each source zone. b value is almost unity in the Source Zone 1,4,5,6,7,9 whereas it is higher in areas 2,3 and lower in Source Zone 8. This means that sufficient data are available in the Zone 1,4,5,6,7,9 and in Zone 2,3,8 either earthquake data are missing or may be due to tectonic reasons.

7. Attenuation relationship specific for Himalyan region have not been developed yet. As Nepal lies in the subduction zone, Youngs et al. 1997, Atkinson and Boore 2003 and Lin and Lee 2008 have been found appropriate for our study. Whereas, Youngs et al. 1997 has been used in this study.
8. Total hazard curve for both rock site and soil site are calculated and plotted.
9. Probabilistic seismic hazard analysis has been carried out to access PGA and SA for 30 sites within Sankhuwasabha district for return period of 475 and 2475 years for rock and soil site condition. PGA at rock and soil sites are 0.54 and 0.74 respectively both for 10% probability of exceedance in 50 years. Similarly, PGA for rock site and soil site with uniform probability of 2% exceedance in 50 years are 0.87 and 1.32 respectively.

5.2 CONCLUSION AND DISCUSSION

The probabilistic seismic hazard analysis is done for the site of Sankhuwasabha District considering all available sources that may produce strong ground motions of engineering significance. Young's attenuation model developed for subduction zone was used for assessment of seismic hazards both at rock and soil sites.

Most of the public, commercial and industrial building in Sankhuwasabha district are being designed and constructed without following proper guidelines for earthquake resistance. This part of the country has equal chance of witnessing damaging earthquakes like other parts. Hence, it cannot be denied that the structures are vulnerable and people are in high risk. In fact, the reliable estimation of seismic hazard and risk eventually minimized social and economic disruption caused by earthquakes. In order to develop the seismic input parameters for the design and construction of earthquake resistant structures in this region the output of this work would be valuable information.

The PSHA Module is used to develop comprehensive catalog of consistent magnitude, G-R relation for each source zone there by develop hazard curve and uniform hazard spectra. The module can help future researchers for seismic hazard analysis for other districts as well. The developed earthquake catalog contains 1124 events after removing and declustering the earthquake data but it was seen that there was less remarkable earthquake near vicinity of our study area.

Delineation of Earthquake source is an important part of hazard estimation. Proper division of available source zone provides relevant results.

Only mainshocks, after declustering of available earthquake catalogue, were considered in Seismic hazard assessment. Whereas, magnitude frequency relation was slightly deviated for some of the sources, resulting some discrepancy in Hazard curves between available literatures.

PGA at rock and soil sites are 0.54 and 0.74 respectively both for 10% probability of exceedance in 50 years period. PGA for rock site and soil site with uniform probability of 2% exceedance in 50 years are 0.87 and 1.32 respectively, which is slightly higher

obtained by (Bhattarai G. , 2010), where PGA at rock and soil sites in the order of 0.29 and 0.39g was estimated in eastern Nepal, both for 10% probability of exceedance in 50 years. For both rock and soil sites the maximum spectral acceleration is observed at time period of 0.1 second. The maximum spectral acceleration at soil site is 2.9g for return period of 2500 years, while it is 1.8g for return period of 500 years. At the rock site maximum spectral acceleration is observed as 1.7 and 1.0 g for return period of 2500 and 500 years respectively. Whereas, PGA(g) at bed rock for return period of 500 years is only 0.1g calculated in Seismic Hazard map (Pandey, et al., 2002). The highest PGA value (i.e. 350 gal) is found to the east i.e. border between Sankhuwasabha and Taplejung districts (Bhattarai et al., 2011).

5.3 RECOMMENDATIONS

Following further studies can be done in future:

1. Himalayan regions have long history of earthquakes, however there is not sufficient data available yet hence attenuation law specific to the region is yet to be established. PGA and SA are very sensitive to attenuation relationships. Therefore, specific attenuation relation that is derived specially for Nepal need to be develop which will provide more reasonable result.
2. No linear sources for earthquake are incorporated in this study resulting underestimation for larger earthquakes and their effects.
3. The use of alternative models based on logic tree can be studied. Each of alternative models can be assigned a weightage factor that is interpreted as the relative likelihood of that model being correct.
4. Mean rate of exceedance can be disaggregated with respect to magnitudes and distance.

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